A Project report on

"CYBER THREATS MONITORING FRAMEOWRK FOR WOMEN SAFETY IN SOCIAL MEDIA PLATFORMS"

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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SANTHIRAM ENGINEERING COLLEGE::NANDYAL

(AUTONOMOUS)

(Approved by AICTE, New Delhi: Affiliated to J.N.T. University, Anantapuramu, A.P.) Accredited by NAAC (Grade-A), Accredited by NBA (ECE&CSE)

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CERTIFICATE

This is to certify that the dissertation entitled "CYBER THREAT MONITORING FRAMEWORK FOR WOMEN SAFETY IN SOCIAL MEDIA PLATFORM" is being submitted by C.MADHU SUDHAKAR (23X55A3202), D.ABHINAI REDDY (22X51A3203), D.LAKSHMI NARASIMHA (22X51A3209), M.SRIHARI (22X51A3232), for partial fulfilment of the award of the Degree of Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE) in the SANTHIRAM ENGINEERING COLLEGE, Nandyal (Affiliated to J.N.T.University, Anantapur) (AUTONOMUS) is a record of bonafied work carried out by them under our guidance and supervision. The results embodied in this project have not been submitted to any other university for the award of any Degree.

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LIST OF ABBREVATION

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
API	Application Programming Interface
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
BERT	Bidirectional Encoder Representations from Transformers
TF-IDF	Term Frequency–Inverse Document Frequency
SVM	Support Vector Machine
NB	Naive Bayes
RF	Random Forest
KNN	K-Nearest Neighbors
GPU	Graphics Processing Unit
CPU	Central Processing Unit
RAM	Random Access Memory
OS	Operating System
UI	User Interface
UX	User Experience
REST	Representational State Transfer
JSON	JavaScript Object Notation
CSV	Comma-Separated Values
URL	Uniform Resource Locator
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDS	Intrusion Detection System
SIEM	Security Information and Event Management
IoT	Internet of Things
SMOTE	Synthetic Minority Over-sampling Technique
F1-score	Harmonic Mean of Precision and Recall
TP	True Positive
FP	False Positive
TN	True Negative

ABSTRACT

The exponential growth of social media platforms has introduced new challenges related to user safety, privacy, and psychological well-being. While these platforms enable global communication and self-expression, they have simultaneously become environments where women are disproportionately exposed to cyber threats such as harassment, cyberbullying, stalking, impersonation, and abusive behavior. Existing safety mechanisms largely depend on manual reporting and static moderation policies, which are insufficient to address the dynamic and large-scale nature of online abuse.

This paper proposes an intelligent Cyber Threats Monitoring Framework designed specifically to enhance women's safety across social media platforms. The proposed system integrates Natural Language Processing (NLP), Machine Learning (ML), and behavioral analytics to monitor online interactions in real time. Textual content such as posts, comments, and messages is analyzed for abusive language, threats, and emotional manipulation using transformer-based NLP models, while behavioral patterns are examined to identify stalking, impersonation, and suspicious activities. Detected threats are classified based on severity, enabling automated alerts and escalation mechanisms to protect users proactively.

Experimental evaluation demonstrates that the proposed framework achieves high accuracy in detecting cyber threats and significantly reduces response time compared to traditional reporting-based systems. The framework offers a scalable, adaptive, and proactive solution that can serve as a foundational model for next-generation digital safety systems aimed at protecting women in online environments.

Key words: Cyber Threats, Women Safety, NLP, Machine Learning, Cyberbullying, Harassment, Stalking, Impersonation.

CHAPTER 1

INTRODUCTION

1.1 Introduction

As social networks like Facebook, Twitter (X), Instagram, Reddit, YouTube continue to expand, the use of internet-based communication is becoming part of our daily lives; however, while allowing for people to connect on a global scale and share information with one another, these same networks create opportunities for users to become victims of various types of cyber threats, such as cyberbullying, harassment, stalking, impersonation, and abusive behaviour. Women are at a significantly higher risk for experiencing the forms of digital harassment outlined above than men.

Presently, traditional systems of moderating content on social media platforms rely primarily upon users reporting content, after which it is reviewed by a human, either via an email or in-person. And, because users report harmful content and must wait for that content to be reviewed by a human reviewer, detection of threatening content takes a great deal of time; therefore, the harmful content may stay live on the social media platform for an extended period of time, resulting in victimisation experiences being of psychological stress, violation of privacy, and threats to safety.

In order to address the issues previously outlined, a Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms will be designed and developed. The Cyber Threat Monitoring Framework will use Artificial Intelligence, Natural Language Processing (NLP), Machine Learning, and Behavioural Analytics to automatically identify harmful content and suspicious user activity in real-time across multiple social media platforms.

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1.2 Objective

This project intends to build an automatic, intelligent system capable of detecting and monitoring cyber threats against women used on social media. The system will utilize machine learning and natural language processing to identify the presence of abusive language, harassment, cyberbullying, impersonation, and suspicious behavior patterns.

The specific objectives of the proposed system are:

- Collect social media data from various sources, including Twitter, Reddit, YouTube, and online news sources, through the use of application programming interfaces (APIs).
- Preprocess and analyze the textual content of social media posts, comments, and tweets using natural language processing techniques.
- Utilize machine learning models trained on lethality and bias for text classification for the automatic determination of harmful or abusive language.
- Classify the threats identified into the appropriate category (e.g., cyberbullying, harassment, impersonation, stalking, or suspicious activity).
- Create and present analytics through interactive dashboards, charts, and other forms of graphical representation.
- Create and distribute automated threat analysis reports that summarize the potential threat, confidence score(s), and other relevant information.
- Provide users with the ability to enter specific keywords into a scalable web-based platform to conduct threat analysis.

The goal of the project will ultimately be to create a proactive monitoring system that provides real-time detection of cyber threats and empowers women to be safer online.

1.3 Artificial Intelligence and Machine Learning

Cyber threats are being detected and analyzed through the help of AI and machine learning on social media platforms. Through giving computers the ability to analyze large amounts of text, they can find patterns that indicate abusive or harmful behavior.

AI and machine learning are used to help systems learn from the data and improve over time (machine-learning) by learning to classify social media content as normal (not threatening) or threatening (with abusive language) based on features of the raw text input.

The system trains models with data sets containing examples of abusive language, cyber bullying, and hate speech. Once trained, these models can automatically identify patterns in social media content and detect threats that may be missed through typical keyword-based filtering.

Additionally, continuously learning from data allows the machine learning models to make the detection process more accurate and adaptable to changing behavior on the internet.

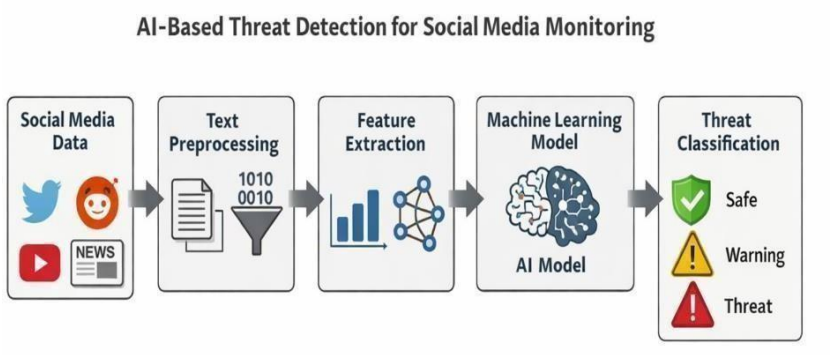

Figure-1.1 Artificial Intelligence Threat Detection Model

1.4 Natural Language Processing

The purpose of Natural Language Processing (NLP) is the study of machine understanding of human language. This allows the computer system's ability to comprehend various aspects of written word content (e.g., posts, comments and tweets) to identify whether they contain threats or otherwise violate policies against abusive behaviours.

For this initiative, natural language processing (NLP) techniques have been applied to accomplish multiple methods such as pre-processing of text, tokenizing into separate words, performing sentiment analysis and identifying keyterms as the means of converting unstructured text into structured format to be subsequently

analysed usingmachine learning algorithms.

In addition to producing structured data from the written content, NLP techniques also allow the systems to understand the contextual meanings of words. This is vital when trying to identify subtle forms of harassment (e.g., sarcasm), emotional manipulation, etc. Furthermore, the combination of NLP techniques and machine-learning algorithms will enable the system to identify trends within harmful language that traditionally keyword-based filters would be unable to identify.

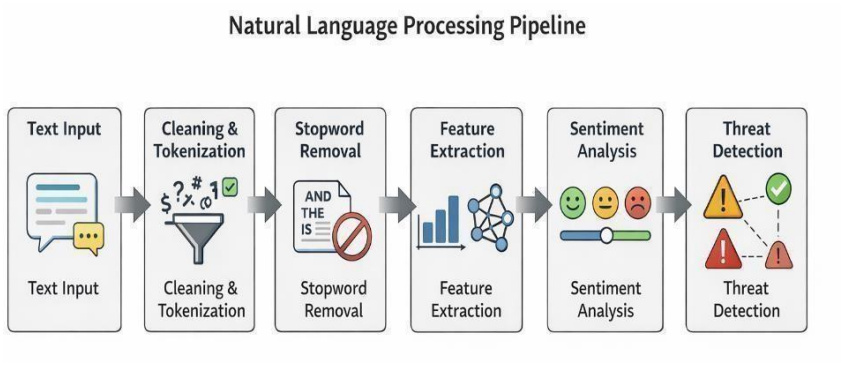

Figure -1.2 Natural Language Processing Pipeline

1.5 Threat Classification

The categorization of harmful digital content that has been detected by a cyber threat is called "threat classification". The proposed solution will categorise all detected cyber threats based on the defining features of both their severity and type.

There are at least six categories of cyber threats that are identified in the proposed system:

1. Cyber Bullying
2. Cyber Harassment
3. Abusive Language
4. Stalking
5. Impersonation
6. Suspicious Behaviour Patterns

A confidence value will be assigned to each detected threat to allow the determination of the likelihood that this content may be harmful. A high confidence value will prompt an email alert to be sent to the appropriate party, as well as create a report for additional investigation.

Threat classification will help users determine what type of threat has been detected and, if necessary, enable law enforcement/administrators to take proper action.

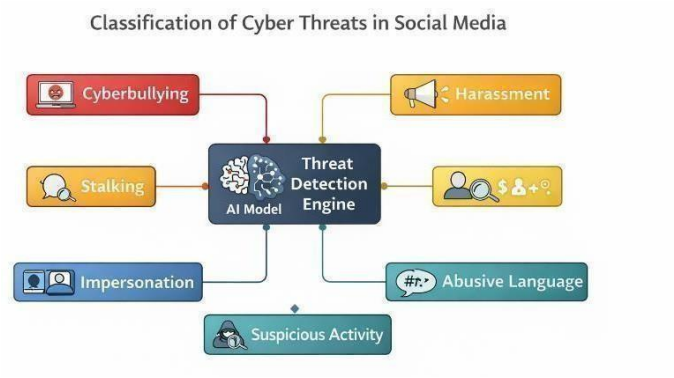

Figure -1.3 Threat Classification Model

CHAPTER 2

LITERATURE SURVEY

2.1 Relate Work

Concern over cyber threats, including online harassment on social media, has grown significantly in recent times. Researchers are investigating ways to identify these types of language usage as well as methods for identifying someone who is being bullied by applying various machine learning (ML) and natural language processing (NLP) techniques.

Originally, researchers identified this type of behaviour using either specific words or phrases (i.e., keyword-based) and applied these filtering techniques to see if the account they were using on social media contained any of these identified words or phrases. Because most filtering solutions use predefined (by the solution's creator) lists with specific words/phrases, they could not identify words/phrases that have a context-based meaning or were similar to "banter" or slang terms (which develop over time) that would be present in the online environment; therefore, the detection of many accounts would not have been successful in identifying harmful behaviour.

Researchers in subsequent studies began taking machine-learning (ML) approaches to identify instances of cyberbullying. Researchers used supervised learning algorithms (such as: Support Vector Machines (SVM), Naive Bayes classifying algorithms, Random Forests - classifiers) to determine the presence of abusive text data from actual social media applications, thus allowing this data to be used to train ML algorithms to determine how to assess text as abusive and not abusive. ML approaches resulted in a significant increase in the accuracy of identifying harmful language compared to keyword-based approaches.

The development of advanced deep learning technologies has led to newer, more accurate ways of identifying and detecting cyber threats through the use of neural network-based approaches. CNNs (Convolutional Neural Networks) automatically extract features from text-based data while RNNs (Recurrent Neural Networks) and LSTMs (Long Short-Term Memory) analyze temporal relationships within textual data. Using these different neural architecture types allows one to identify not only the context of a word but also the relationships between words as well as how to detect and define different forms of online harassment.

The latest research in this area has focused primarily on using better-performing transformer-based

architectures. Examples include BERT (Bidirectional Encoder Representations from Transformers), RoBERTa, and GPT-based architectures. These types of neural networks leverage attention mechanisms to identify sentence context, resulting in improved accuracy over current methods of identifying hate speech, abusive language, or cyberbullying. Thus far transformer models have been successful at analyzing social media datasets that have large amounts of text and subtle signs of malicious behaviour.

Researchers have used various methods to detect suspicious activities like stalking, impersonation, or coordinated campaigns via behavioral analysis, in addition to text analysis. Behavioral analysis looks at things like how often a person messages an individual (repeated messaging), how many times they interact with a user (abnormal interaction frequency), how they interact with their followers (unusual follower activities), and when there was an abrupt change in their behavior. The combination of behavioral analytics with text analytics can help identify much more complex threat patterns than are able to be identified with text analytics.

Although there have been recent developments in algorithms for detecting cyber threats, most of the existing systems are based on detecting specific cyber threats (e.g., detecting cyberbullying or hate speech). Few systems have been able to create an integrated framework that enables the simultaneous monitoring of multiple social media platforms and allows for an analysis of both textual content and behavioral patterns. As such, the Cyber Threat Monitoring Framework for Women's Safety on Social Media Platforms will provide an integrated solution to these existing limitations by utilizing Natural Language Processing (NLP), Machine Learning, and Behavioral Analysis.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Workflow

The Cyber Threat Monitoring framework for Women Safety consists of an automated means of monitoring and analyzing social media content to identify potential forms of cyber hazards, including harassment, cyber bullying, harmful language, stalking and impersonation. Monitoring social media data is performed through multiple stages, which include collecting the social media data, preprocessing the data, analyzing the threats, classifying the threats and visualizing the results to the user at the end.

The entire workflow of the proposed Cyber Threat Monitoring System (CTM) is comprised of the following five phases:

1. **Data Collection**

Collects Textual Data (e.g., posts, comments, tweets, articles, etc.) using APIs from multiple social media platforms. In addition to Twitter, Reddit and YouTube, the following news services will be monitored; GNews, NewsAPI, etc...

2. **Data Pre-Processing**

Data that was previously collected often has noise (e.g., special characters, emojis, links, etc.) that will require removal before the system can analyze it for threats. Therefore, during pre-processing, the system will strip undesirable data out of the text, convert all text to lowercase, tokenize the text into individual words, and remove any stop-words.

3. **Feature Extraction**

Use NLP techniques to extract pertinent features from the purged data. These features will be used to find patterns that are representative of abusive or threatening behavior.

4. **Machine Learning for Detecting Cyber Threats**

The machine learning model culls the analyzed data to find malicious material within it. The sentimental value, toxicity level, and contextual meaning of each piece or collection of text are used by the models to assess if the text contains cyber-threats.

5. **Classifying Cyber Threats**

Once a cyber threat has been identified, the system assigns it to a pre-set category. Examples of category types include: cyberbullying, harassment, abusive language, stalking, and suspicious activity.

6. **Visualizing and Reporting on Detected Cyber Threats**

The results of the machine learning model's output are presented via web-based interactive interface. This includes charts, graphs, and summary statistics. Detailed PDF reports can also be generated that provide information on the threat that was detected, along with the associated confidence score and other information and data.

This process is effectively configured to allow users to effectively and efficiently detect cyber-threats while providing users with the ability to track and monitor real-time online activities and analyze these activities in real-time.

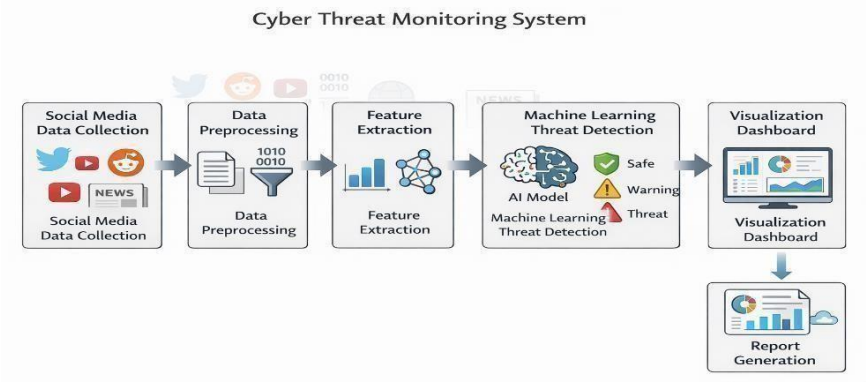

Figure -3.1 Workflow of Cyber Threat Monitoring System

3.1.1 Data Sources

The Various online platforms provide data that can be used by the proposed system using public API's to collect text based on keywords to help identify potential for Cyber Threats through a large volume of user-generated text.

There is a wide variety of sources that will be available to the proposed system as follows:

- Twitter **:** Tweets & Comments containing the User's postings based on specified keywords
- Reddit **:** Discussion Posts & Comments made throughout the various sub-reddits.
- YouTube **:** Video Titles, Descriptions, and Comments made in the Comments Section.
- News APIs **:** Articles and Headlines collected from sites like GNews and NewsAPI

The Digital Platforms listed above provide us with Real-Time Data to analyze for harmful or abusive language. The information collected in this system includes Username, Post Content, Time Stamp, and URL Links.

The use of multiple Digital Platforms to identify the Threats allows us to identify them across the various Digital Environments therefore we are building a more complete monitoring system.

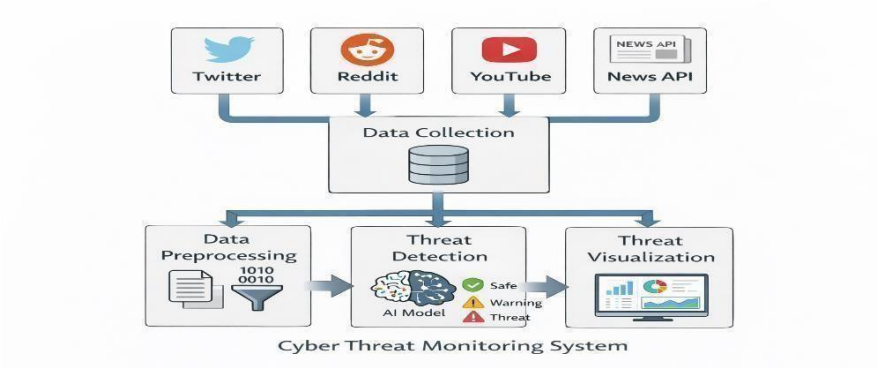

Figure -3.2 Social Media Data Sources

3.1.2 Data Pre-processing

Image Performing Data Preprocessing has become a vital component in preparing Social Media Website (Social Media) raw content for analysis. Many pieces of text found on social media contain "NOISE"; that is, there are a large variety of different elements like URLs, hashtags, emojis, special character content and irregularities of format that exist within the text which need to be cleaned to allow use of that text for effective analysis.

The operations performed during the data Preprocessing stage are as follows:

- **Text cleaning :** Removal of symbols, punctuation marks, and URLs
- **Lowercase conversion :** Conversion of characters to lowercase to maintain uniformity
- **Tokenization :** Dividing the body of the text into separate and smaller pieces or groups of size 1 (or "token" [1 word]); these token(s) can also consist of a group/sets/combination of token(s) (i.e., "multi word" tokens).
- Removal of "stop words" such as the, is, and, etc.

These data preprocessing methods will greatly reduce any noise present in the data set, thereby improving the machine learning algorithms used for detection and identification of potential threats.

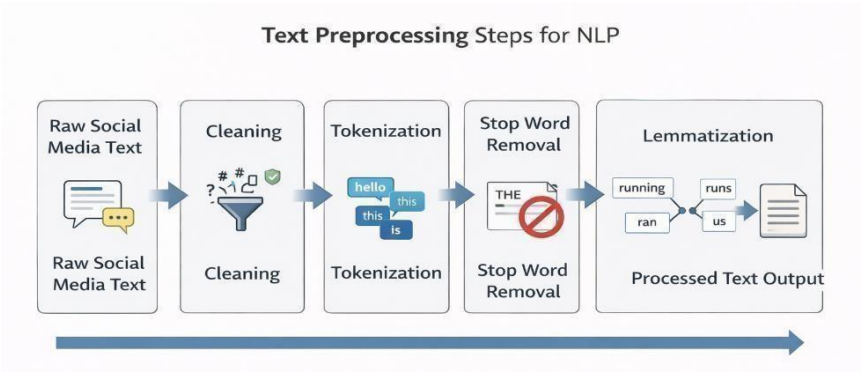

Figure-3.3 Text Preprocessing Pipeline

3.1.3 Threat Detection

Oncepre-processed, text fromthecleaningprocedureis analyzed via MLtodiscoverrisks. Aspects ofthe analysis include: toxic/positive scores, abuse keywords, and context-based meaning.

Usingan MLclassification algorithm with historical data, we are able to compute thelevel of threat in a given piece of text bypredicting the level of risk associated with each element found in that text.

Thetool will allow you not onlyto identifya "threat"that is obvious in nature, but alsoless obvious threats in the form of sarcasm, emotional abuse/manipulation, indirect intimidation etc.

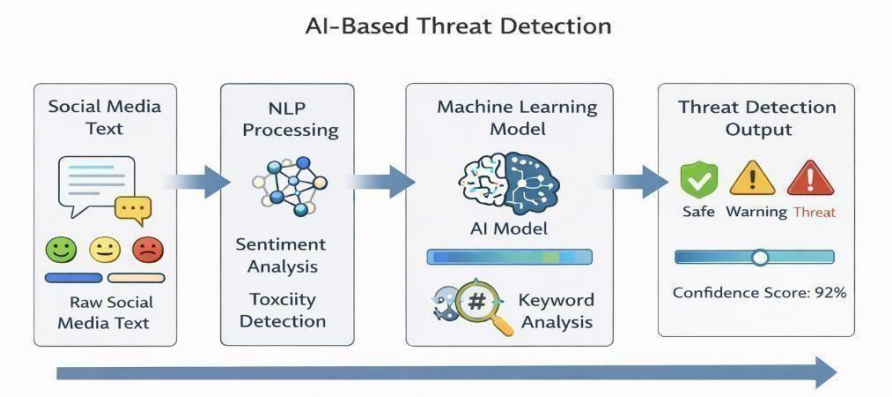

Figure-3.4 Threat Detection Process

3.2 UseCase Diagram

A use case diagram is a graphical representation used in software engineering to describe the interactions between users (actors) and the system. It helps in understanding how a system operates from the user's perspective by identifying the different functionalities provided by the system and the ways in which users interact with those functionalities. Use case diagrams are widely used during system design to visualize system behavior, define system boundaries, and illustrate the relationships between users and system processes.

In the proposed Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms, the use case diagram illustrates how users interact with the system to detect and analyze potential cyber threats present across various social media platforms. The diagram identifies the primary actor, which is the user**,** and demonstrates the different operations that the user can perform through the system interface.

The primary goal of the system is to allow users to monitor social media platforms for harmful or abusive content related to specific keywords or topics. The interaction begins when the user accesses the web-based application and enters a set of keywords related to a particular topic, individual, or potential cyber threat. These keywords may include terms related to cyberbullying, harassment, stalking, abusive language, or other suspicious online activities that may pose risks to individuals, particularly women.

Once the user provides the keywords, the system initiates the **data collection process**. In this stage, the system communicates with multiple social media platforms through their respective APIs, such as Twitter API, Reddit API, YouTube API, and news aggregation APIs like GNews and NewsAPI. The system retrieves relevant content including posts, comments, tweets, video descriptions, and news articles that match the specified keywords. This process allows the system to gather a large volume of real-time social media data for analysis.

After collecting the data, the system performs **text preprocessing operations** to clean and prepare the data for analysis. This stage involves removing unwanted characters, URLs, emojis, and special symbols that may interfere with the analysis process. The text is then converted into a standardized format using techniques such as tokenization, stop-word removal, and normalization. These

preprocessing steps ensure that the textual data is structured and suitable for machine learning-based analysis.

Once preprocessing is completed, the system proceeds to the **threat analysis phase**, which is the core functionality of the proposed framework. In this stage, Natural Language Processing (NLP) and machine learning algorithms are applied to analyze the textual data and identify patterns associated with harmful behavior. The system evaluates factors such as sentiment polarity, toxicity levels, presence of abusive keywords, and contextual meaning of sentences. Based on these features, the system determines whether the content contains potential cyber threats.

If harmful or suspicious content is detected, the system performs **threat classification**, where the identified threats are categorized into predefined classes. These categories may include cyberbullying, harassment, abusive language, stalking behavior, impersonation attempts, and other suspicious activities. Each detected threat is assigned a confidence score that represents the likelihood that the content is harmful or threatening. Higher confidence scores indicate stronger evidence of potential cyber threats.

After the threat detection and classification processes are completed, the system presents the results to the user through an interactive **visualization dashboard**. The dashboard provides a clear representation of the analysis results using charts, graphs, and statistical summaries. Users can view information such as the number of threats detected, the total number of posts scanned, and the distribution of threat categories. These visualizations help users quickly interpret the results and identify areas of concern.

In addition to visual representations, the system also provides the option to generate **detailed threat reports**. These reports summarize the findings of the analysis and include important information such as the detected threat content, username or source of the post, platform name, timestamp, keywords identified in the text, and the confidence score of the detection. The reports can be exported in document format and used for documentation, monitoring, or further investigation.

The use case diagram therefore illustrates the complete interaction between the user and the system, including actions such as entering keywords, collecting social media data, performing threat detection, displaying visualization results, and generating detailed reports. It highlights the major functionalities of the system and demonstrates how the cyber threat monitoring framework assists users in identifying and analyzing harmful online activities.

By clearly depicting the relationship between the user and the system components, the use case diagram helps in understanding the overall functionality of the proposed framework. It also serves as a valuable tool for system design and documentation, as it outlines the major processes involved in detecting and monitoring cyber threats across social media platforms.

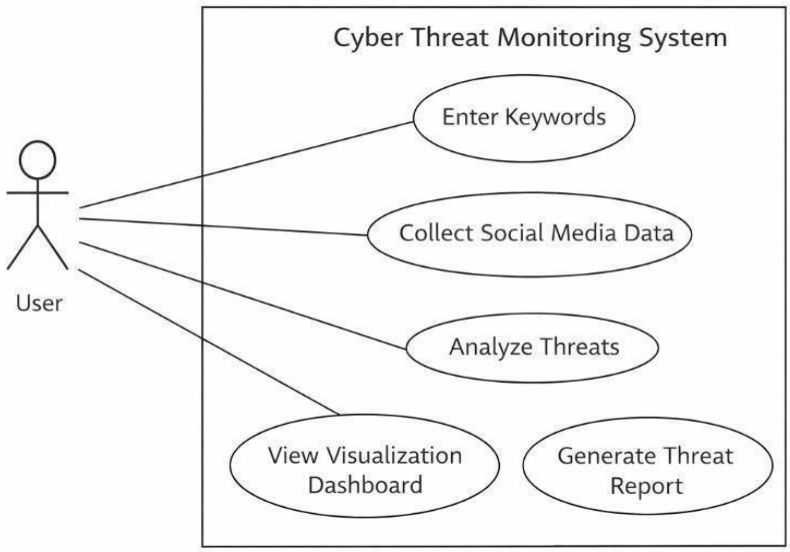

Figure -3.5 Use Case Diagram

3.3 Control Flow Diagram

A control flow diagram is used to illustrate the sequence of operations that occur within a system and how data moves through different stages of processing. It helps represent the logical flow of actions performed by the system from the beginning of a task to its final output. Control flow diagrams are useful for understanding how a system processes inputs, performs intermediate operations, and produces results. They also help developers and users visualize the internal functioning of the system and identify how different components interact with one another.

In the proposed **Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms**, the control flow diagram describes the step-by-step process used by the system to detect potential cyber threats from social media platforms. It represents how data flows from the user input stage to the final threat detection and reporting stage.

The process begins when the **user enters one or more keywords** into the system through the web-based interface. These keywords may represent a specific topic, event, person, or suspicious activity that

the user wants to monitor. The keywords act as search queries that allow the system to identify relevant content from multiple social media platforms.

Once the keywords are submitted, the system initiates the **data collection phase**. During this phase, the system retrieves posts, comments, tweets, video descriptions, and news articles from various social media platforms such as Twitter, Reddit, YouTube, and online news sources using their respective APIs. This collected data forms the raw dataset that will be analyzed for potential threats.

After collecting the data, the system performs **data preprocessing operations** to prepare the textual information for analysis. This stage involves cleaning the text by removing unnecessary characters such as URLs, emojis, special symbols, and punctuation marks. The text is also converted into a standardized format through processes such as tokenization, stop-word removal, and normalization. These preprocessing steps ensure that the data is structured and ready for further analysis.

Following preprocessing, the system applies **Natural Language Processing (NLP) techniques** to analyze the textual content. NLP methods allow the system to understand the meaning and context of the language used in social media posts. The system examines factors such as sentiment polarity, presence of abusive words, emotional tone, and contextual patterns that may indicate harmful behavior.

Once the textual features are extracted, the processed data is passed to the **machine learning threat detection model**. The model analyzes the features and determines whether the content contains potential cyber threats. The system evaluates each piece of content and assigns a confidence score indicating the likelihood that the content is harmful or threatening.

If the system identifies potentially harmful content, it moves to the **threat classification stage**. In this stage, the detected threats are categorized into predefined threat categories such as cyberbullying, harassment, abusive language, stalking behavior, impersonation attempts, or suspicious activity. This classification helps users understand the nature and severity of the detected threat.

After classification, the system generates **visualization results and reports**. The detected threats are displayed on the user dashboard through charts, graphs, and detailed information cards. These visualizations provide users with a clear understanding of the number of threats detected, the type of threats, and their corresponding confidence scores.

Finally, the system generates a **detailed threat report**, which summarizes the analysis results. The report includes important information such as the detected threat content, the platform from which the content was retrieved, the associated username, timestamps, keywords identified, and the confidence score of the detection. This report can be downloaded or stored for documentation and further investigation.

The control flow diagram therefore represents the complete operational flow of the cyber threat monitoring system, showing how data moves from user input through various processing stages until the final detection results are presented. It provides a clear overview of the internal functioning of the system and helps users and developers understand how the system processes social media data to detect potential cyber threats.

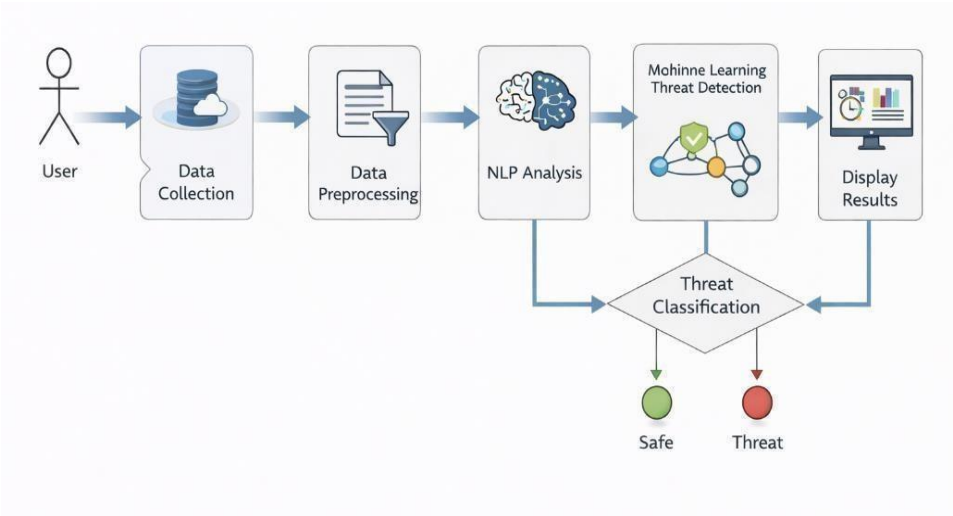

Figure-3.6 Control Flow Diagram

3.4 Deployment Diagram

A deployment diagram is used to represent the physical architecture of a system and illustrate how various software components are deployed across hardware environments. It shows the relationship between system nodes such as servers, databases, client devices, and external services. Deployment diagrams help visualize how different parts of a system communicate with each other and how the application is distributed across physical or virtual infrastructure.

In the proposed **Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms**, the deployment diagram describes the physical arrangement of the system components and

the communication between them. The system architecture is designed using a client–server model where users interact with the application through a web interface, while the backend server performs the core processing tasks including data collection, threat analysis, and report generation.

The system consists of three main components: the **client-side interface, the backend server, and external social media services**.

The first component is the **front-end web-based user interface**, which runs on the user's device through a web browser. This interface allows users to interact with the system by entering keywords related to potential cyber threats. The user interface is developed using web technologies such as HTML, CSS, and JavaScript, which provide an interactive and user-friendly environment. When a user submits a request, the interface sends a request to the backend server using REST API calls. These API requests contain information such as the entered keywords and other parameters needed for the analysis.

The second component is the **backend server**, which acts as the central processing unit of the system. The backend server receives requests from the user interface through the REST API. Once a request is received, the server performs several operations including data collection, preprocessing, threat detection, and threat classification. The server retrieves relevant content from various social media platforms by communicating with external APIs such as Twitter API, Reddit API, YouTube API, and news aggregation APIs like NewsAPI or GNews.

After collecting the data, the backend server processes the textual information using Natural Language Processing techniques and machine learning models. These models analyze the data to identify patterns associated with cyber threats such as harassment, abusive language, stalking, or suspicious behavior. Once the analysis is completed, the server sends the results back to the front-end interface.

The third component is the **visualization and reporting module**, which displays the results of the threat analysis to the user. The results are presented through a web-based dashboard that includes charts, statistics, and detailed threat information. Users can view detected threats, confidence scores, and other relevant metadata. The system also provides the option to generate comprehensive reports summarizing the detected threats.

The deployment architecture of the system is designed to support **scalability and efficient performance**. By separating the client interface from the backend processing server, the system can

handle multiple user requests simultaneously. The backend server can process large volumes of data and communicate with external APIs without affecting the responsiveness of the user interface.

This deployment structure enables the system to function effectively in real-world environments by supporting multiple users, processing large datasets, and integrating with various social media platforms. The deployment diagram therefore provides a clear representation of how the cyber threat monitoring system is distributed across different components and how these components interact to perform threat detection and monitoring.

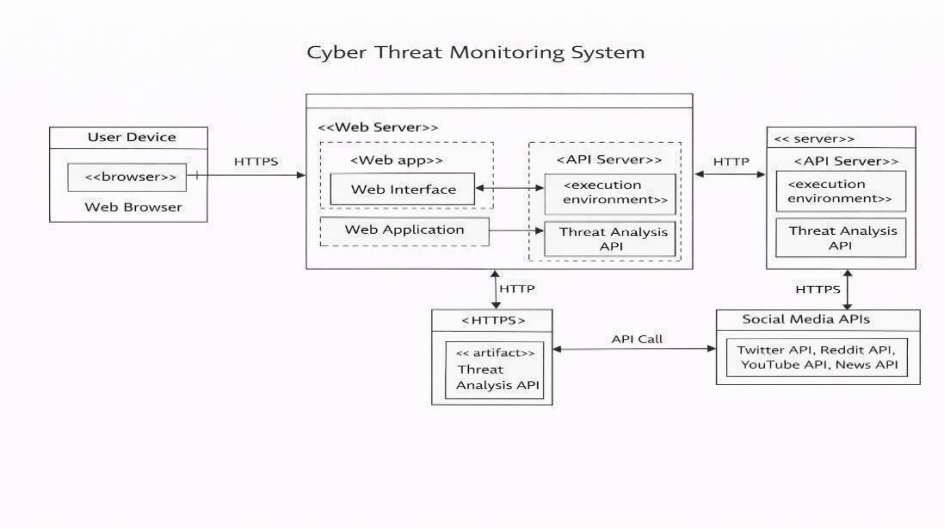

Figure -3.7 Deployment of System

3.5 Sequence Diagram

A sequence diagram is a type of Unified Modeling Language (UML) diagram that illustrates how different components of a system interact with each other over time. It shows the order in which messages are exchanged between system components and helps visualize how a process is executed step by step. Sequence diagrams are particularly useful for understanding the flow of control within a system and the interactions that occur between users, interfaces, servers, and external services.

In the proposed **Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms**, the sequence diagram represents the communication between the major system components during the threat detection process. It demonstrates how the user, frontend interface, backend server, social media APIs, and machine learning model interact with each other to analyze social media data and detect potential cyber threats.

The process begins when the **user enters one or more keywords** into the web-based application interface. These keywords may represent topics, names, or phrases associated with potential cyber threats such as harassment, abusive language, or suspicious activities. Once the user submits the keywords, the request is sent from the **frontend component**, also known as the client-side interface, to the **backend server** through a REST API request.

Upon receiving the request, the backend server initiates the **data retrieval process**. During this stage, the backend server communicates with multiple social media platforms using their respective APIs, such as Twitter API, Reddit API, YouTube API, and news APIs. The server sends queries to these platforms using the keywords provided by the user and retrieves relevant content including posts, comments, tweets, and article descriptions.

After collecting the data from these external services, the backend server proceeds with the **data processing phase**. The retrieved data is first preprocessed to remove unnecessary characters, links, and irrelevant text elements. Natural Language Processing (NLP) techniques are then applied to extract meaningful features from the textual content.

Following preprocessing, the processed data is passed to the **machine learning threat detection model**. The machine learning model analyzes the textual data to determine whether the content contains harmful or abusive language. The model evaluates several factors such as sentiment polarity, toxicity levels, presence of abusive keywords, and contextual meaning. Based on this analysis, the model identifies potential cyber threats and assigns confidence scores indicating the likelihood that the content is harmful.

Once the threat detection process is completed, the backend server prepares the **analysis results**, which may include detected threats, confidence scores, threat categories, and relevant metadata such as usernames and timestamps. These results are then sent back to the frontend interface as a response to the original request.

Finally, the frontend interface receives the response and **displays the results to the user** through an interactive dashboard. The dashboard presents the detected threats using visual elements such as cards, charts, and statistical summaries. Users can also view detailed information about each detected threat and generate comprehensive reports if required.

The sequence diagram therefore illustrates the entire communication process between the system components, starting from user input and ending with the presentation of threat detection results. It clearly shows how the frontend interface, backend server, external APIs, and machine learning model work together to analyze social media data and detect potential cyber threats. By representing these interactions in a chronological order, the sequence diagram helps developers and users understand the internal workflow of the cyber threat monitoring system.

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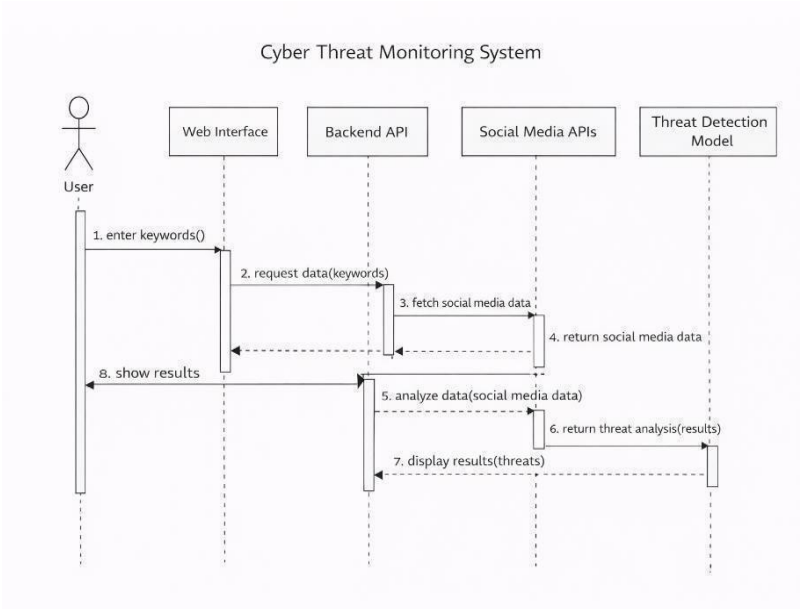

Figure -3.8 Sequence diagram

3.6 Class Diagram

A class diagram is one of the most important diagrams in the Unified Modeling Language (UML) used to represent the structural design of a system. It illustrates the different classes within the system, along with their attributes, methods, and the relationships that exist between them. Class diagrams help developers understand how the system is organized and how different components interact with each other to perform various tasks.

In the proposed **Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms**, the class diagram represents the major components of the system and the relationships between them. Each class in the system performs a specific function that contributes to the overall operation of the cyber threat monitoring framework.

The main classes involved in the system include **User Interface (UI), Data Collector, Text Processor (TP), Threat Detection Model (TDM), Threat Classifier (TC), and Report Generator (RG)**. These classes work together to collect social media data, analyze potential threats, classify detected threats, and present the results to the user.

The **User Interface (UI)** class acts as the interaction layer between the user and the system. It provides the web-based interface through which users can enter keywords related to potential cyber threats. The UI class also displays the results of the threat analysis, including detected threats, confidence scores, and visualizations such as charts and dashboards. Additionally, it allows users to generate detailed reports based on the detected threats.

The **Data Collector** class is responsible for retrieving relevant data from multiple social media platforms. It communicates with external APIs such as Twitter API, Reddit API, YouTube API, and news APIs to collect posts, comments, tweets, and articles that match the keywords entered by the user. The collected data is then forwarded to the text processing component for further analysis.

The **Text Processor (TP)** class performs preprocessing operations on the collected textual data. This includes tasks such as removing special characters, eliminating URLs, converting text to lowercase, tokenization, stop-word removal, and normalization. These preprocessing steps ensure that the text data is clean and structured before it is analyzed by the machine learning model.

The **Threat Detection Model (TDM)** class is the core component responsible for identifying harmful or abusive content within the processed data. This class uses machine learning and natural language processing techniques to analyze textual features such as sentiment, toxicity levels, contextual meaning, and keyword patterns. Based on this analysis, the model determines whether the content contains potential cyber threats.

Once a threat is detected, the processed information is passed to the **Threat Classifier (TC)** class. This component categorizes the detected threats into different types such as cyberbullying, harassment, abusive language, stalking behavior, impersonation attempts, or other suspicious activities. The classification process also assigns confidence scores to indicate the severity or likelihood of the detected threat.

The final component is the **Report Generator (RG)** class. This class compiles the analysis results and generates detailed reports that summarize the detected threats. The report includes information such

as the content of the detected threat, the platform from which it was retrieved, the associated username, timestamps, detected keywords, and confidence scores. These reports can be viewed by the user through the system interface or downloaded for documentation and further investigation.

The relationships between these classes represent the workflow of the system. The User Interface interacts with the Data Collector to initiate data retrieval. The Data Collector sends the collected data to the Text Processor for preprocessing. The processed data is then analyzed by the Threat Detection Model, and the detected threats are categorized by the Threat Classifier. Finally, the results are passed to the Report Generator, which produces the final output displayed to the user.

The class diagram therefore provides a clear representation of the internal structure of the cyber threat monitoring system and highlights how different components collaborate to perform threat detection and monitoring effectively.

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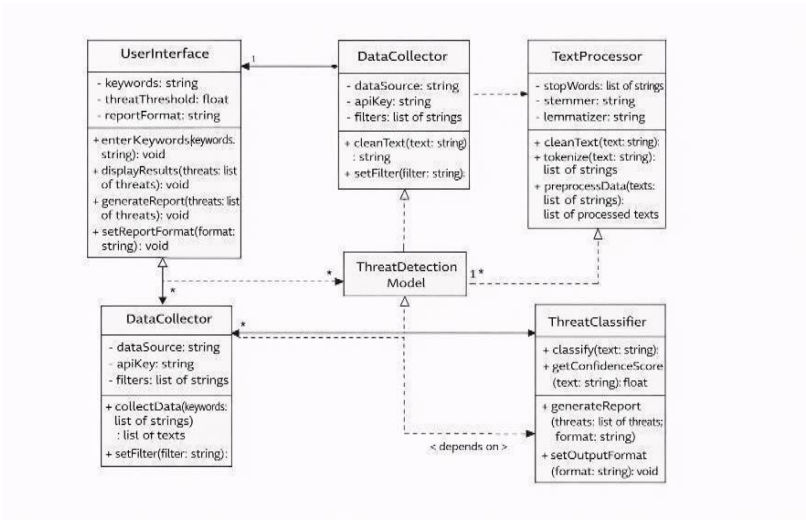

Figure 3.9 Class diagram

3.7 Activity Diagram

An activity diagram is a type of Unified Modeling Language (UML) diagram used to represent the workflow of a system. It describes the sequence of activities and actions that occur from the beginning of a process to its completion. Activity diagrams are helpful in visualizing the dynamic behavior of a system by showing how different activities are connected and how the control flows from one activity to another. They also help in understanding the overall operational flow of the system and identifying the decision points within the process.

In the proposed **Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms**, the activity diagram illustrates the step-by-step workflow followed by the system to detect cyber threats from social media data. It represents the flow of activities starting from user input and ending with the presentation of threat detection results.

The process begins when the **user accesses the web-based application** and enters one or more keywords related to potential cyber threats. These keywords may include specific names, phrases, or topics that the user wants to monitor on social media platforms. After entering the keywords, the user submits the request through the application interface.

Once the request is submitted, the system initiates the **data collection activity**. During this stage, the system communicates with various social media platforms such as Twitter, Reddit, YouTube, and online news sources through their respective APIs. The system retrieves posts, comments, tweets, and articles that match the user-provided keywords.

After collecting the relevant data, the system proceeds to the **data preprocessing activity**. In this stage, the raw textual data is cleaned and prepared for analysis. Preprocessing tasks include removing unnecessary characters such as URLs and emojis, converting text to lowercase, tokenizing the text into individual words, and removing common stop words. These steps help improve the quality of the data and make it suitable for machine learning analysis.

Following preprocessing, the system performs the **threat analysis activity**. In this stage, Natural Language Processing techniques and machine learning models are applied to analyze the textual content. The system examines the sentiment of the text, identifies toxic or abusive language, and detects patterns that may indicate cyberbullying, harassment, or other forms of harmful behavior.

Once the analysis is completed, the system moves to a **decision point** where it determines whether any potential threats have been detected. If no threats are identified, the system displays a message indicating that no harmful content was found for the given keywords. If threats are detected, the system proceeds to the **threat classification activity**, where the detected threats are categorized into predefined types such as cyberbullying, harassment, abusive language, stalking, or suspicious activity.

After classification, the system generates **visual representations and detailed reports**. The results are displayed on the dashboard using charts, graphs, and threat cards that provide information such as

confidence scores and detected keywords. The system also generates a comprehensive threat analysis report that summarizes the detected threats along with relevant metadata.

Finally, the process ends with the **display of the analysis results to the user**, allowing the user to review the detected threats and download the generated reports if necessary.

The activity diagram therefore provides a clear representation of the entire workflow of the cyber threat monitoring system, showing how the system processes user input, performs threat analysis, and generates the final output. It helps in understanding the operational flow of the system and how different activities are coordinated to detect potential cyber threats effectively.

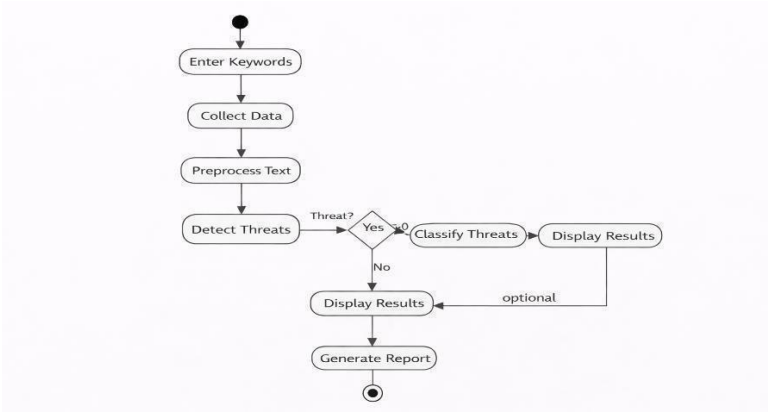

Figure 3.10 Activity diagram

CHAPTER4

SOFTWARE& HARDWARE REQUIREMENTS

4.1 Introduction

System System requirements specify the minimum amount of hardware and software needed for an application to run successfully. System requirements are established in order to guarantee that the system functions correctly, maximizes its output efficiently, and has no limitations imposed by its hardware or software. The establishment of correct system configurations is very important when you are implementing machine learning applications and web applications.

The proposed Cyber Threat Monitoring Framework for Women's Safety in Social Media Platforms requires a combination of hardware resources and software tools that will be used for tasks such as collecting data, preprocessing text, machine learning analysis, creating visual representations, and generating reports.

The proposed system is web-based and consists of a front-end interface for interaction with users and a back-end server (using Natural Language Processing and Machine Learning) that performs threat analysis. Therefore the system will require enough processor speed, RAM, and a sufficient number of software libraries (for example: Python, R, Tensor Flow) to perform the data processing and live threat detection.

4.2 Hardware Requirements

Hardware The hardware requirements consist of the physical hardware required to run the system efficiently; taking into account the fact that the proposed system performs data analysis and ML operations, it requires a system with sufficient processing and memory resources.

The following are the recommended hardware specifications for use in implementing the system:

- Processor: Intel Core i5 or greater Processor - Minimum 2.0 GHz processor speed
- RAM: Minimum 8GB RAM (to help ensure a system operation with minimal issues associated with the machine learning tasks)
- Storage: Minimum of 500GB HD (for storing project files; cover images; datasets; and generated reports)
- video graphics card (Optional): NVIDIA GPUs (such as GTX 1650 or greater) to increase speed of machine learning computations and model training

- Internet connection: Stable Internet connection to access social media APIs and to retrieve real-time data.

Byproviding these hardware components, the system will be able to effectively process large volumes of social media data and perform the threat detection process more efficiently.

4.3 Software Requirements

The required software components are listed below.

- Code editor - Visual Studio Code

  Visual Studio Code (VS Code) is the code editor used to develop this document. It is a lightweight, feature-rich source-code editor that supports many programming languages. Developers can debug their code, view syntax highlighting, use version control integration with Git, and extend VS Code with plug-ins to create their own code development environments.

- Operating systems - Windows / Linux

  Both Windows and Linux operating systems can be used to run the application. Both operating systems have ample support for ML libraries and software development frameworks that will be used.

- Programming language - Python

  Python is the main programming language that will be used for the project back end. It has powerful libraries to support machine-learning, natural-language processing, and data analysis. It has simple-to-understand syntax and a large community of users/developers to help support the creation of AI-based applications.

- Back End Framework - Flask

  A lightweight web framework for developing the system's backend server using the Python programming language is called Flask. Flask serves as a backend server to manage API requests sent to the frontend from the HTML pages on the web browser. API requests processed by the backend will eventually have an output sent back to the frontend user by way of Flask.

Aside from functioning as a backend server, Flask also provides a mechanism to integrate machine learning models with external APIs, allowing full development of web applications that incorporate artificial intelligence (AI).

- Front End Technologies - HTML / CSS / JavaScript

The user interface within the system is developed with HTML, CSS, and JavaScript as follows:

- HTML for establishing the structure of the webpage.
- CSS for providing design and style features to the UI.
- JavaScript to provide dynamic functionality to the UI, e.g., charts, visualizations, and interface interaction via APIs.

These technologies allow for an interactive experience between users and the threat detection data returned via API.

- Machine Learning & Natural Language Processing Libraries

The following python libraries are used for machine learning and text data analysis:

- TensorFlow/PyTorch for constructing and training machine learning models.
- Scikit-learn for applying classification algorithms.
- NLTK/SpaCy for performing natural language processing tasks like tokenization/stemming and determining sentiment.
- Pandas & NumPy for performing data processing/analysis.

- Libraries Used for Visualization

The system uses visualization tools to create graphics. This includes the following libraries:

- Chart.js - used to display data about confidence scores and reported threats.
- Three.js - used to create interactive three-dimensional representations of threats.
- Vanta.js - used to add background animations to the user interface.

- APIs Used to Collect Data

  The system collects real-time information about online activity from various external sources via the following APIs:

  - Twitter API - to gather tweets that contain specified keywords.
  - Reddit API - to pull together relevant posts and comments from the Reddit discussion board.
  - YouTube API - to analyze video titles and comment sections.
  - NewsAPI/GNews API - to collect news articles related to any possible threat.

  These APIs provide the data necessary to analyze online activity and to identify cyber threats.

CHAPTER5

SYSTEM IMPLEMENTATION

5.1 Architecture

The Cyber Threat Monitoring Framework for Women's Safety on Social Media Platforms is a modular architecture that integrates multiple components of social media data collection, text processing, machine learning analysis, and visual representation.

The system is composed of two distinct layers:

1. Frontend Layer: The frontend supports a web-based interface that allows users to interact with the system and enter keywords of concern related to possible cyber threats. The frontend displays graphical representations (charts and visualizations) and detailed documentation of detected threats in response to user input.

2. Backend Processing Layer: All of the major processing tasks take place in the backend including, but not limited to, collecting data, preprocessing text, analyzing through machine learning and classifying threats via reports. The backend layer interacts with various social media application program interfaces (APIs) to retrieve real time data and utilize Natural Language Processing methods to analyze the retrieved data.

This architecture will allow the user to monitor multiple social media platforms concurrently and permit easy scalability of the system as data volume grows.

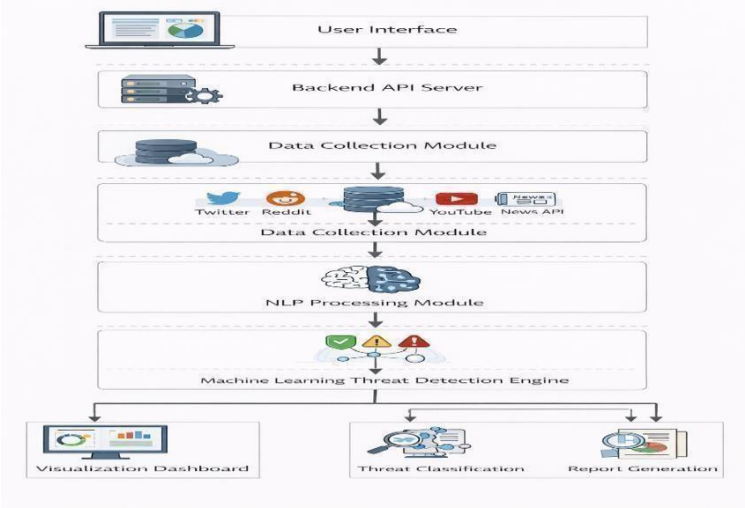

Figure -5.1 System Architecture

5.2 Data Preprocessing

After retrieving social media data from the social media platforms, machine learning models will analyze the collected data after they have been processed. The data processing step converts the raw text data into a suitable format for analysis.

The processing steps are:

- Removal of noise - Removing unnecessary items such as URLs, emojis, special characters and punctuation.
- Lowercase conversion - To standardize data for uniformity all text is converted to lower case.
- Tokenisation - Breaking the text down into individual words or tokens.
- Removal of stop words - The removal of most frequently used words (stop words) in the English language (for example, the, and, a, very, etc.) does not add any significant value to the overall meaning of the text.
- Lemmatization - Converting each individual word in the text to its native base word will result in higher accuracy when performing analytic functions on the data.

Removing these types of noise from the data set will result in better performance from the threat detection model.

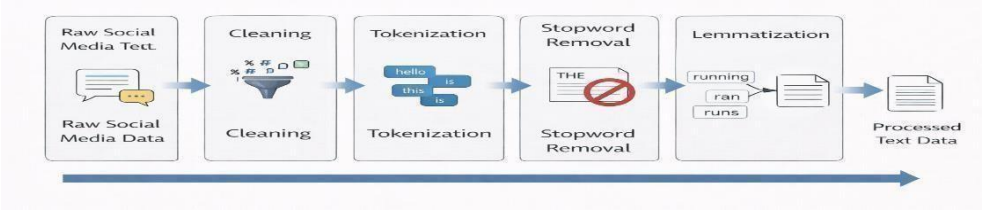

Figure -5.2 Text Processing Pipeline

5.3 Threat Detection Model

The main element of the technology is referred to as the Threat Detection Model. This component utilizes various machine learning and natural language processing methods to identify abusive or harmful content within social media data.

The Threat Detection Model analyzes different text properties such as negative sentiment, occurrence of abusive keywords, toxicity scoring, and context to decide whether there is a potential

cyber threat associated with the content.

In addition to processing text properties, the model is trained on a dataset of examples of abusive and non-abusive messages to learn patterns related to negative behaviors and how to identify them in new social media data.

Once a threat has been identified, the model assigns a confidence value to that threat identifying the likelihood the message contains harmful content. A greater level of confidence denotes a higher likelihood that the identified content contains threatening or abusive language.

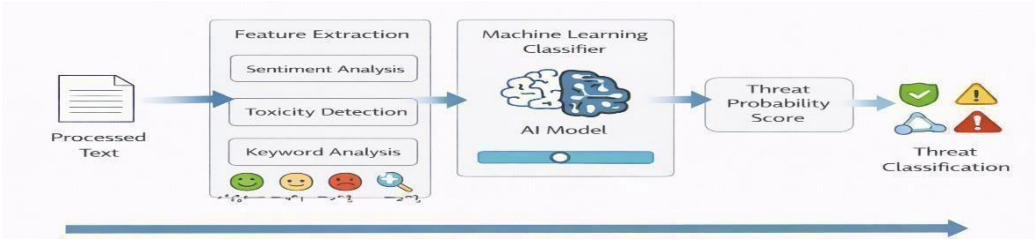

Figure 5.3 Threat Detection Model

5.4 Threat Classification

When there's an identifiable risk, the system first assigns it to one or more of the pre-established classification categories. This creates a better context for users to grasp what kind of threat has been identified and how best to respond.

The core classification groupings are defined as follows:

- Bullying via computers
- Harassment online
- Use of inappropriate language
- Attempts to stalk someone
- Attempts at pretending to be somebody else
- Questionable behaviour

For each risk that has been identified, the system stores metadata, including user name, platform source, date/time/preview of content. In addition, the system also calculates a confidence level for the threat, as to its severity.

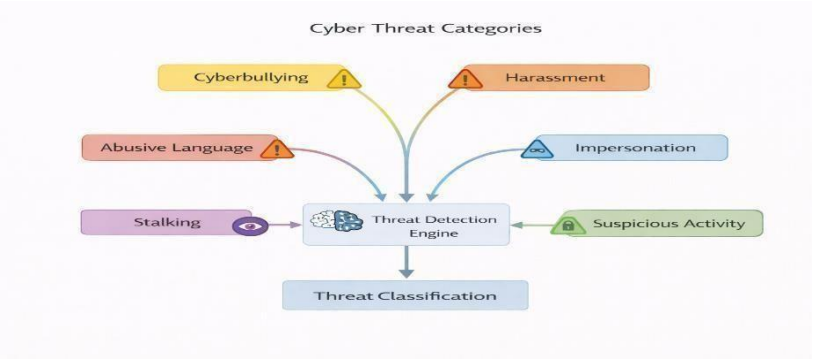

Figure -5.4 Threat Classification Process

5.5 Flow Chart of System

This flowchart shows how the user interacts with the system, entering keywords that initiate a multi-step process to identify threats and generate reports.

The first step in this process is user input (keywords) into the system. After the user submits their keyword(s), the system connects to various social media APIs to collect relevant data (within seconds). Then it preprocesses that data so it can be analyzed byits machine learning (ML) model to detect threats.

Once a threat has been detected by the ML model (prereq. ML detection model must be trained, etc.), the system defines the type of threat (defined by categories) and displays the results on an interactive dashboard. Finally, the user can generate a detailed report summarizing all identified threats.

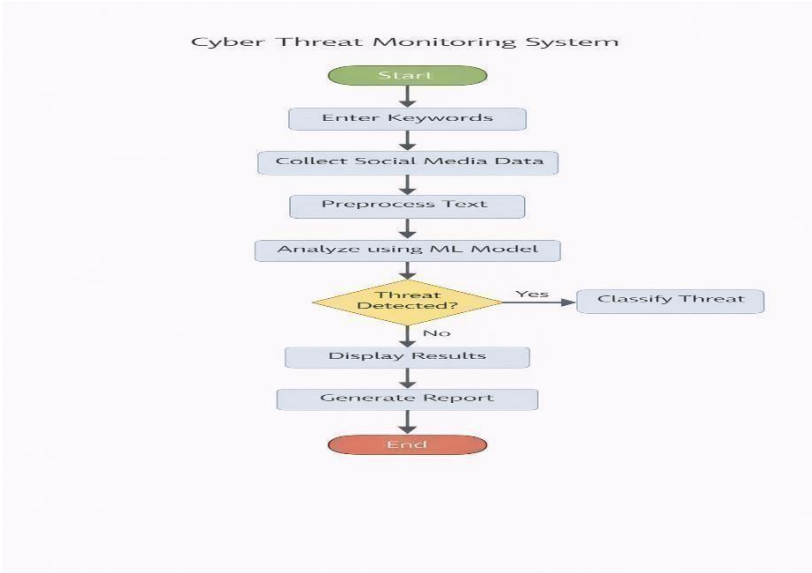

Figure-5.5 Flowchart of Cyber Threat Monitoring System

5.6 Metrics for Evaluation

There are a number of metrics that will be used to measure the performance of the threat detection system:

- Accuracymeasures the number of correctlyidentified (classified) examples as a percentage of the total number of predictions the model made.
- Precision measures the number of correctly identified 'true positives' as a percentage of the total number of positive predictions made by the model.
- Recall measures the number of correctlyidentified 'true positives' as a percentage of the total number of actual 'true positives' (the model's 'ground truth').
- The F1 Score (the harmonic mean of precision and recall) provides a balanced measure of the model's performance.

The above metrics help determine how well the machine learning model is at accurately identifying cyber threats.

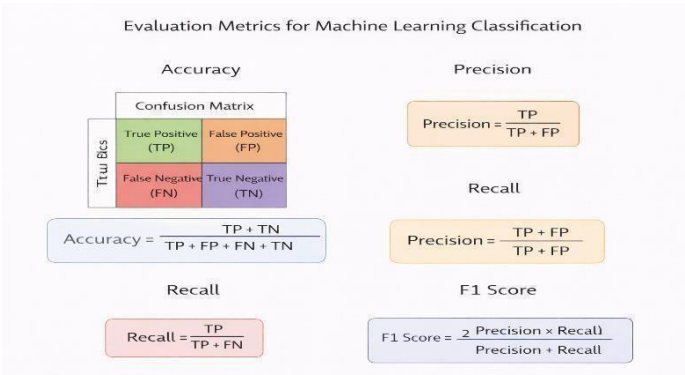

Figure -5.6 Evaluation Metrics

5.7 SOURCECODE

This chapter presents the core source code structure used to implement the Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms. The system is implemented using Python for backend processing and HTML, CSS, and JavaScript for the frontend interface. The backend server is developed using the Flask framework and communicates with the frontend through REST APIs.

The source code consists of multiple modules responsible for different tasks such as data collection, preprocessing, threat detection, visualization, and report generation.

A. Importing Required Libraries

The implementation of the system requires several Python libraries for data processing, machine learning, and API communication.

```
from flask import Flask, request, jsonify
import requests
import pandas as pd
import numpy as np
import nltk
import re
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
```

These libraries are used for handling API requests, processing textual data, and building machine learning models for threat detection.

B. Flask Application Setup

The backend server is created using the Flask framework. Flask enables communication between the frontend interface and the machine learning model.

```
app = Flask(__name__)

@app.route('/api/health', methods=['GET'])
def health():
    return jsonify({"status": "API running"})
```

This route verifies whether the API server is running correctly.

C. Social Media Data Collection

The system retrieves social media data using APIs such as Twitter API, Reddit API, and News APIs.

```
def fetch_news(query):
    url = f"https://newsapi.org/v2/everything?q={query}&apiKey=YOUR_API_KEY"
    response = requests.get(url)
    return response.json()
```

The retrieved data contains text content that will be analyzed bythe threat detection model.

D. Text Preprocessing Function

Before analyzing the data, the text must be cleaned and preprocessed.

```
def preprocess_text(text):
    text = text.lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"[^a-zA-Z ]", "", text)
    tokens = nltk.word_tokenize(text)
    return " ".join(tokens)
```

This function removes unwanted characters, URLs, and punctuation while converting the text into a clean format suitable for machine learning analysis.

E. Threat Detection Model

A machine learning model is used to classifythe text as harmful or non-harmful.

```
vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(dataset['text'])
y = dataset['label']

model = LogisticRegression()
model.fit(X, y)
```

The TF-IDF vectorizer converts textual data into numerical features that can be used bythe classification model.

F. Threat Prediction

Once the model is trained, it can predict whether new text contains harmful content.

```
def predict_threat(text):
    processed = preprocess_text(text)
    vector = vectorizer.transform([processed])
    prediction = model.predict(vector)
    return prediction
```

The system assigns a label to the input text indicating whether it contains potential cyber threats.

G. API Endpoint for Threat Detection

The backend provides an APIendpoint that receives a keyword from the frontend and performs threat analysis.

```
@app.route('/api/analyze', methods=['GET'])
def analyze():
    query = request.args.get('query')
    data = fetch_news(query)

    results = []

    for article in data['articles']:
        content = article['title']
        prediction = predict_threat(content)

        if prediction == 1:
            results.append({
                "content": content,
                "source": article['source']['name']
            })

    return jsonify(results)
```

This endpoint retrieves social media data, analyzes the content, and returns detected threats to the frontend interface.

H. Frontend API Request

The frontend interface communicates with the backend using JavaScript.

```
async function analyzeKeywords() {
  const query = document.getElementById("searchInput").value;

  const response = await fetch(`/api/analyze?query=${query}`);
  const data = await response.json();

  console.log(data);
}
```

This function sends the keyword entered bythe user to the backend API and retrieves the analysis results.

I. Threat Visualization

The system uses JavaScript libraries such as Chart.js to visualize threat detection results.

```
const chart = new Chart(ctx, {
  type: 'bar',
  data: {
    labels: ['Low Threat', 'Medium Threat', 'High Threat'],
    datasets: [{
      label: 'Threat Confidence',
      data: [30, 50, 80]
    }]
  }
});
```

Visualization helps users easily understand the threat distribution and confidence levels.

J. Report Generation

The system can generate a detailed report containing detected threats and analysis results.

```
function generateReport(data) {
  const report = JSON.stringify(data);
  console.log(report);
}
```

Reports can be exported in PDF format for documentation and further investigation.

CHAPTER6

RESULTS & DISCUSSIONS

6.1 Results

Implemented and tested in real time using a variety of social media sources (Twitter, Reddit, YouTube and news APIs), the Cyber Threat Monitoring Framework for Women's Safety (CTMFWSM) evaluated its performance based on its capacity to accurately identify harmful content; classify threats associated with those identified threats; as well as display the results of analysis in a manner that is easily understood by users (via a user-friendly interface).

The final results of the tests indicated that the CTMFWSM is capable of processing large amounts of text (data) within seconds to detect potentially damaging material including, but not limited to, cyberbullying, harassment, and abusive language. Additionally, the CTMFWSM can provide visualization dashboards displaying statistics associated with threats, threat confidence scores, and detection rates, as well as offer users with simple and concise representations of their identified threats.

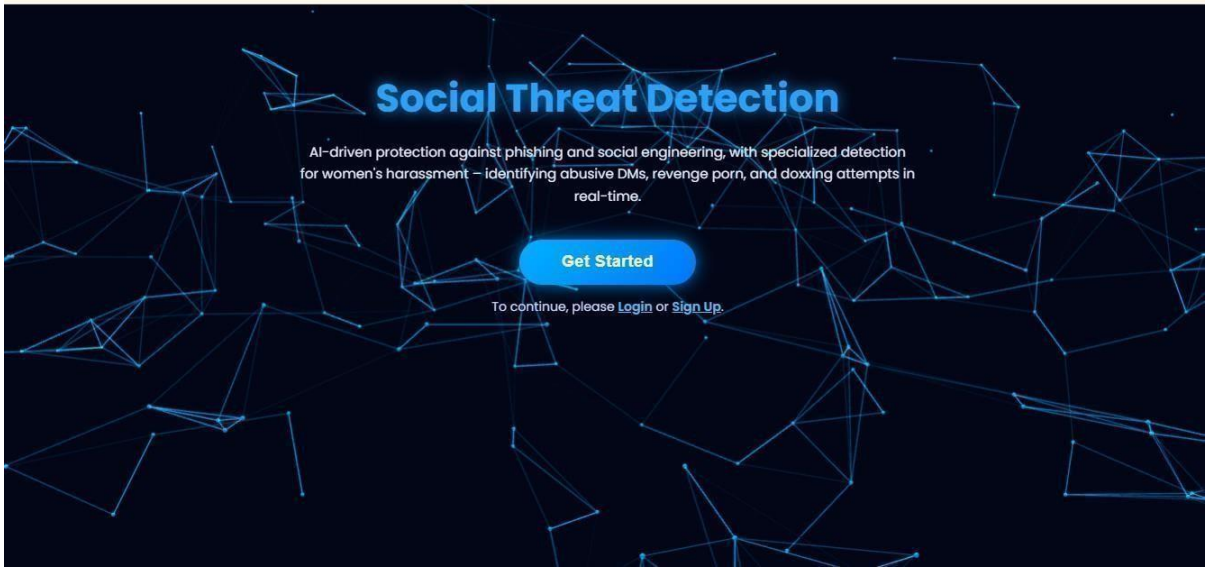

Figure 7.1 Home Page

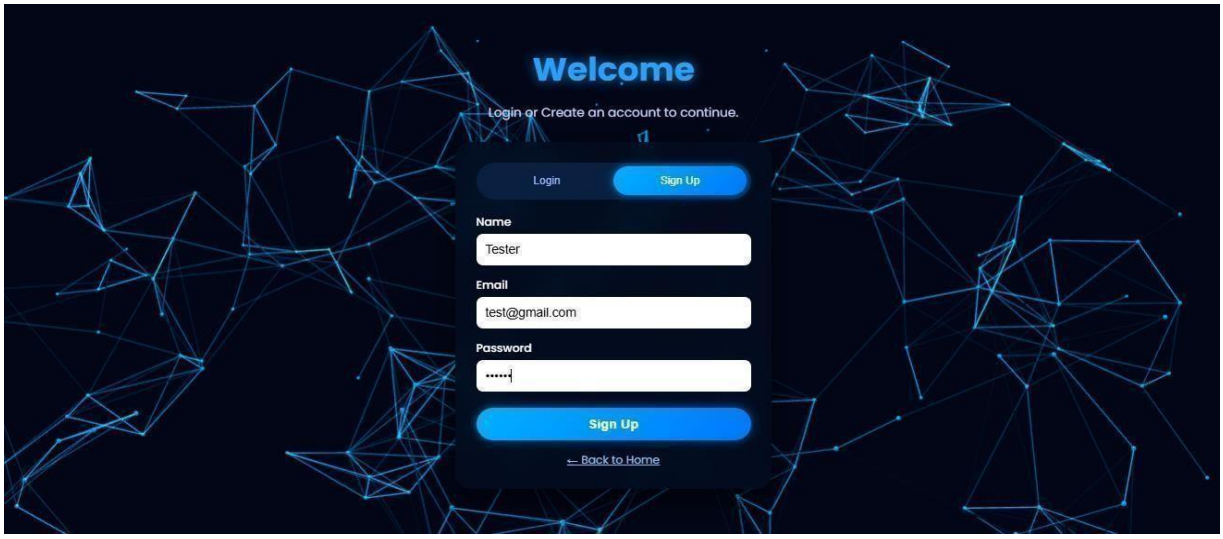

Figure 7.2 Sign Up Page

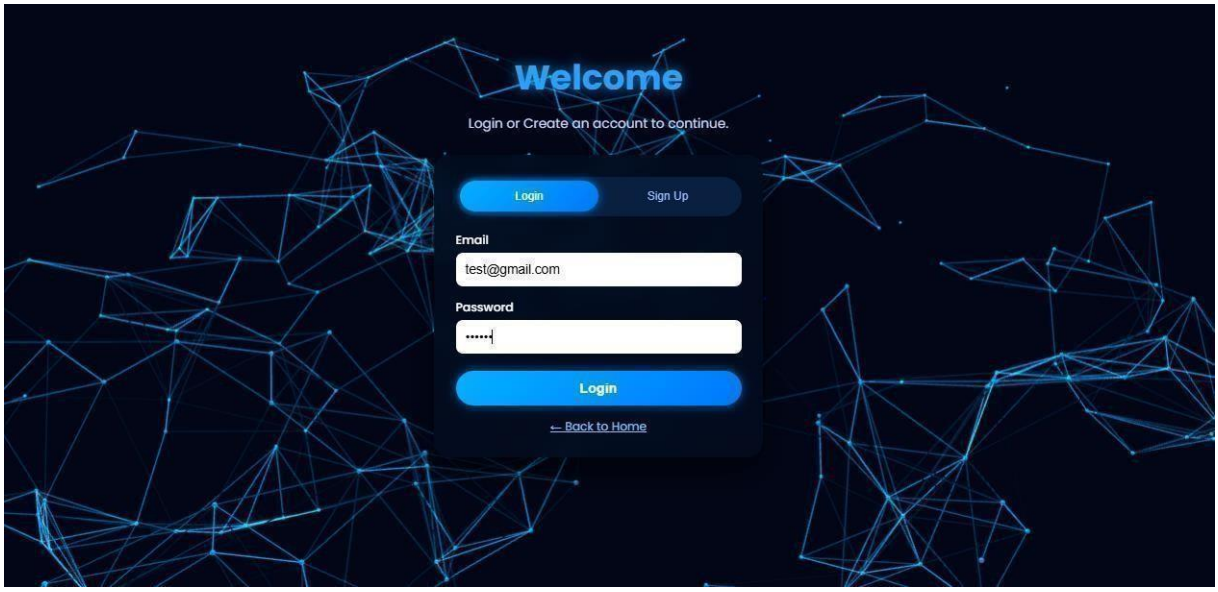

Figure 7.3 Login Page

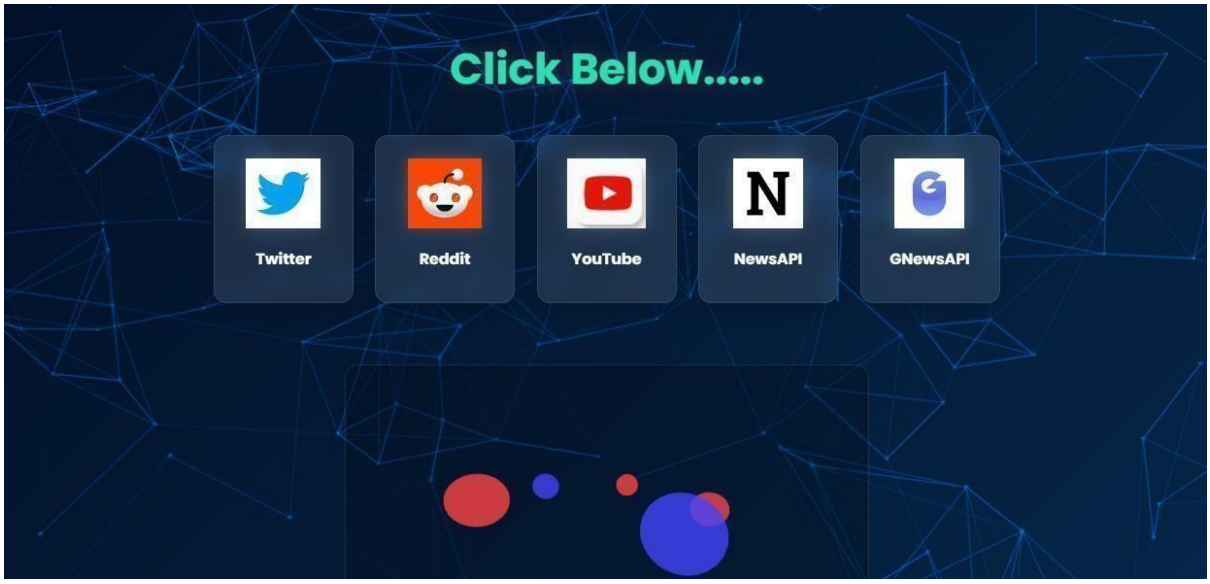

Figure 7.4 Social Media Sites

Test Case 1: Reddit Threat Analysis

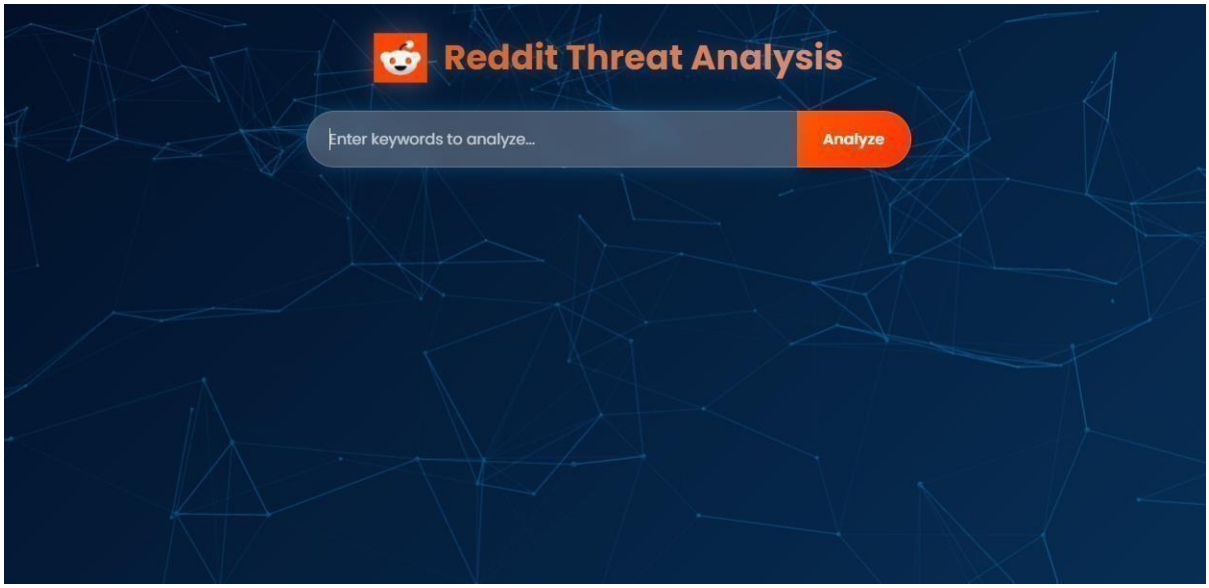

Figure 7.5 Keyword Search Interface

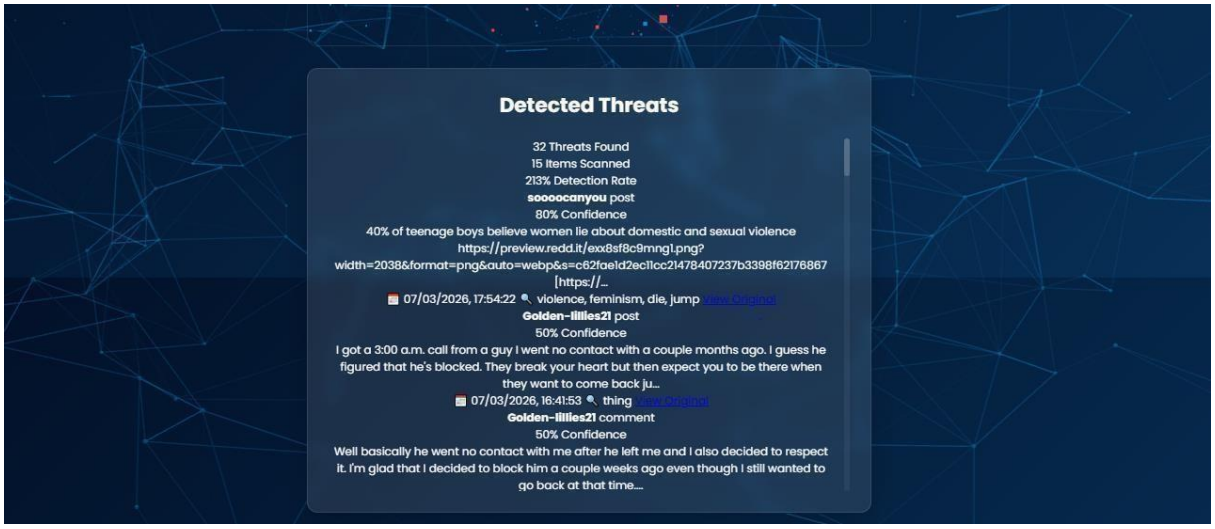

Figure 7.6 Reddit Threat detection results

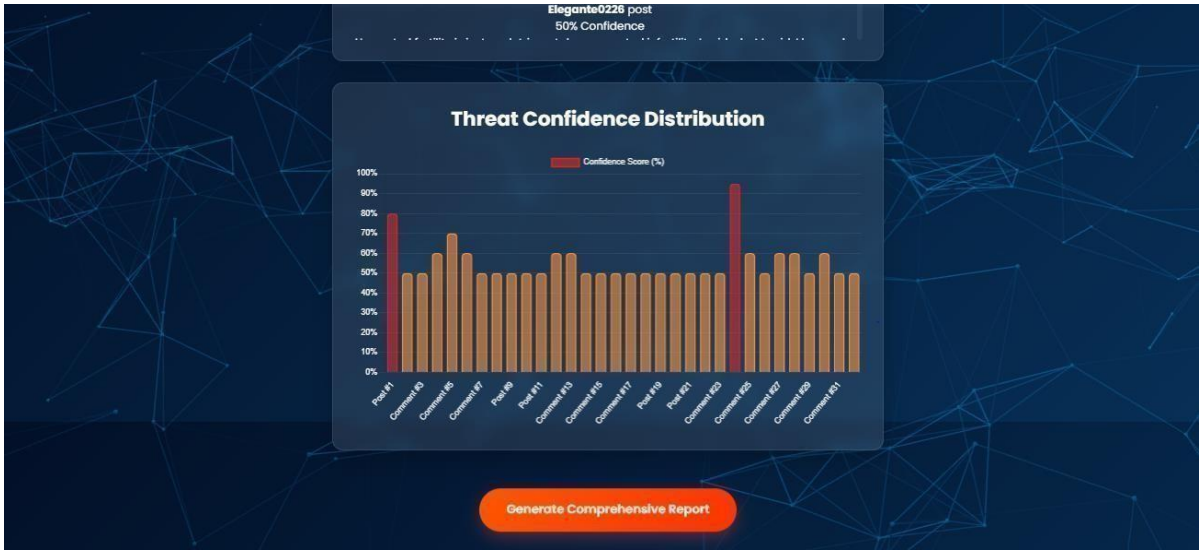

Figure 7.7 Reddit Threat cards with confidence score

Test Case 2: YouTube Threat Analysis

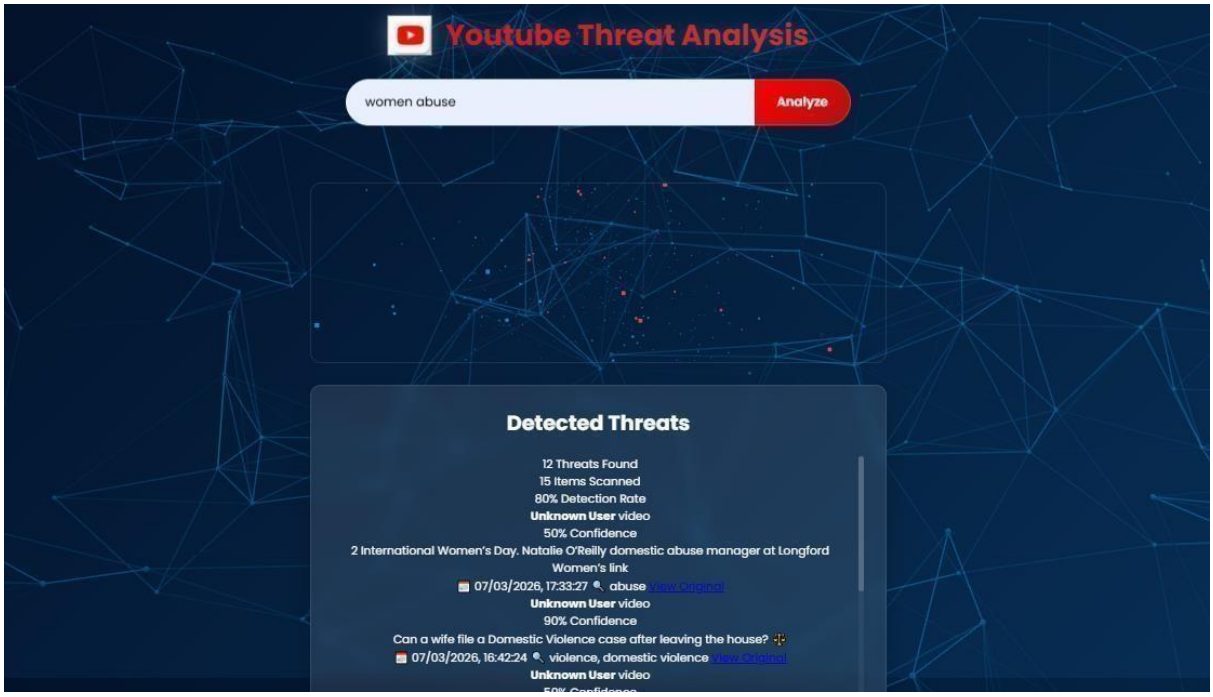

Figure 7.8 YouTube Threat detection results

Figure 7.9 YouTube Threat cards with confidence score

Test Case 3: NewsAPI Threat Analysis

Figure 7.10 NewsAPI Threat detection results

Figure 7.11 NewsAPI Threat cards with confidence score

Test Case 4: GNEWSAPI Threat Analysis

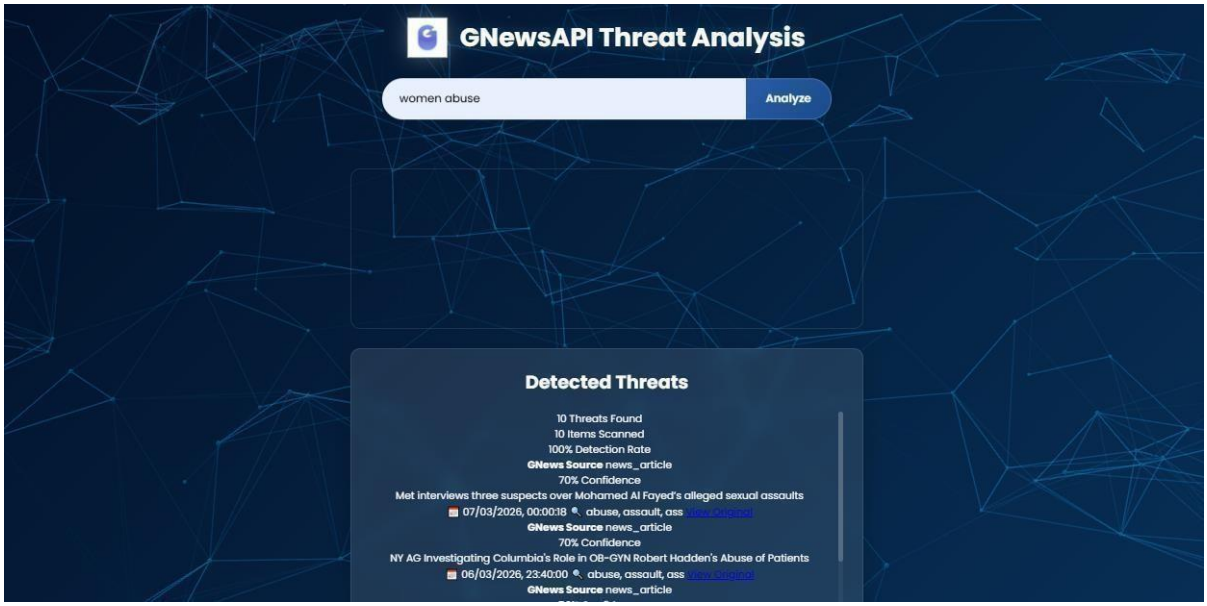

Figure 7.12 GNewsAPI Threat detection results

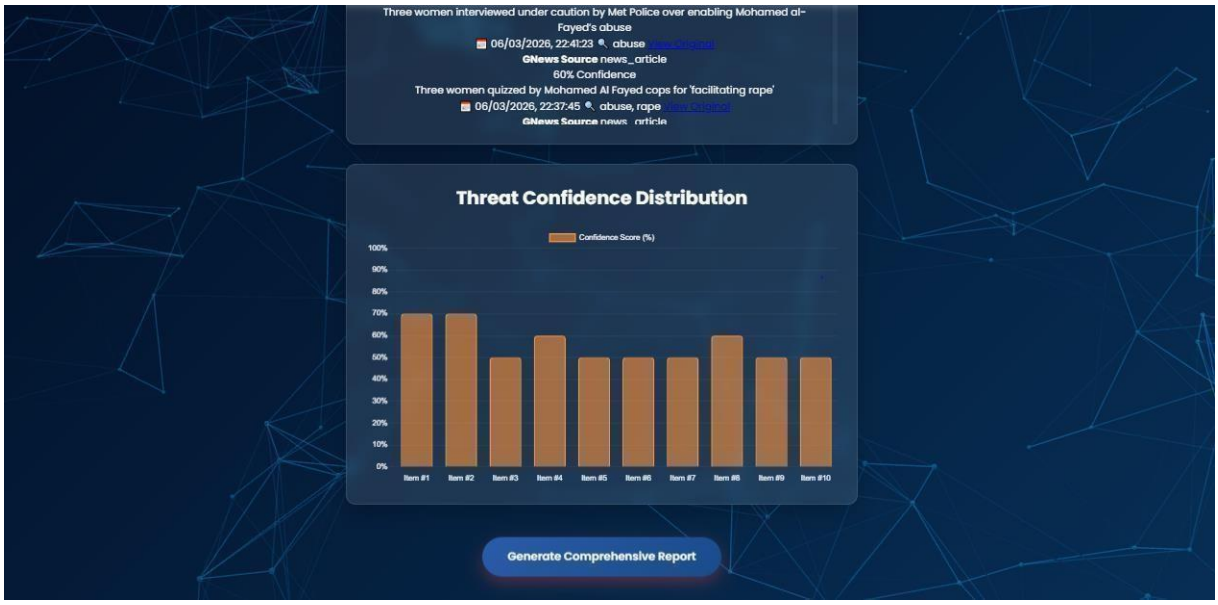

Figure 7.13 GNews APIThreat cards with confidence score

Test Case 5: Twitter Threat Analysis

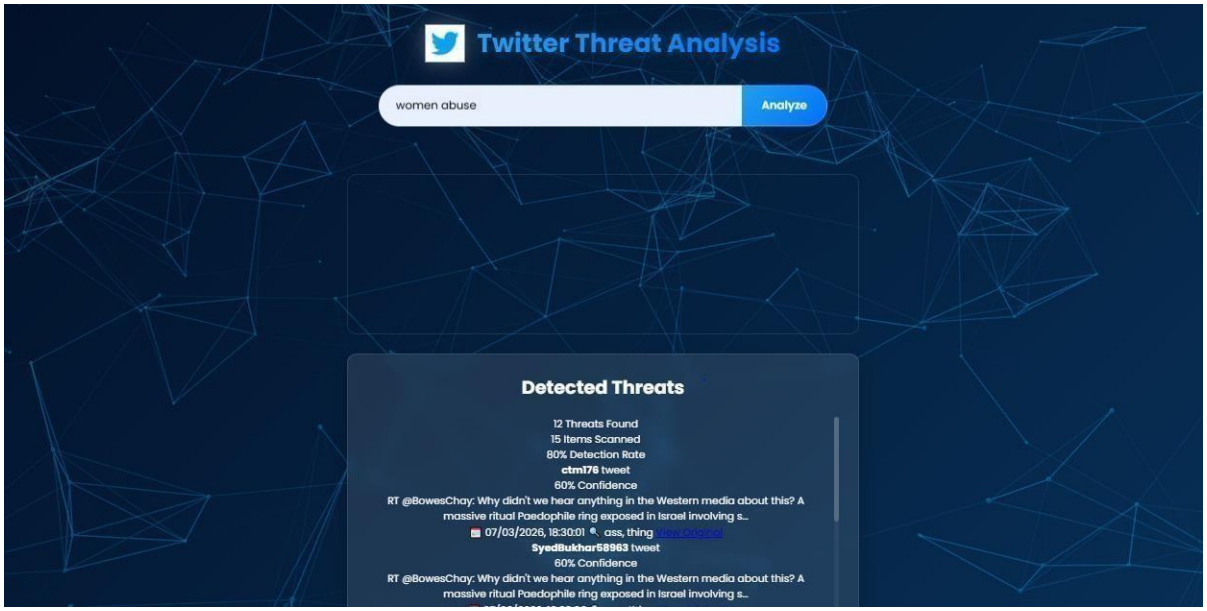

Figure 7.14 Twitter Threat detection results

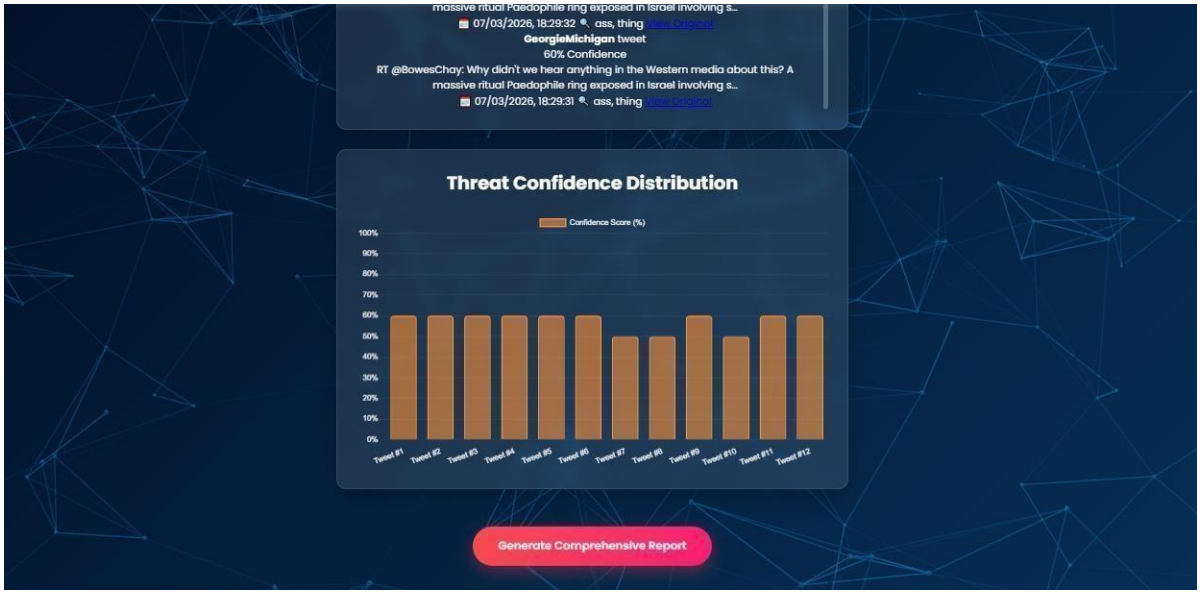

Figure 7.15 Twitter Threat cards with confidence score

6.2 Discussions

The analysis provided by the implemented system, can show that it will allow for real-time monitoring of social media sites for any potential threats to women.

The use of Natural Language Processing and Machine Learning techniques enable analysis of text data for detection purposes; thus allowing for the identification of abusive language, bullying, or any other potentially harmful pattern. Through the use of threat classification, threats that have been detected within the system can be categorized into a preset classification scheme; providing users with a greater understanding of what type of threat exists, as well as how severe a threat may be depending on how many "hits" appeared.

Users will be able to visualize their data through charts and other forms of graphical representation; making interpretation of results much easier. Users also get detailed reports through the generation of a PDF; which provide detailed documentation of detected threats and may assist in assisting researchers and/or via legal analysis.

One of the most beneficial features of this software system is its ability to aggregate data from multiple social media sources at once. In combination with utilizing multiple platforms for monitoring, the system can provide a more comprehensive framework for detection of threats across many different digital environments at the same time.

The quality of training datasets to establish a baseline along with the effectiveness of machine learning models will prove to be an important component in determining if this software system will be successful or not; Continuous improvements on the system's learning model via additional training data, and utilising new and improved Natural Language Processing techniques, will enhance the system's performance.

Overall, the developed framework demonstrates how Artificial Intelligence and Data Analytics (AI/DA) can potentially work in conjunction with each other to increase the safety of individuals (in this case, women) through better monitoring and detection processes.

CHAPTER7

CONCLUSION& FUTURE SCOPE

7.1 Conclusion

With the rise of various social media platforms comes an increase in challenges concerning online safety and digital harassment (specifically for women). There has also been a heightened risk to women due to cyber threats (these include: cyberbullying, harassment, stalking, impersonating, and abusive behaviour). Most social media networks rely upon traditional content moderation systems (manual reporting & intervention) that do not allow for timely detection of threats or incidents.

To help provide an intelligent solution for monitoring & assessing social media interactions using Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP), the Cyber Threat Monitoring Framework for Women Safety in Social Media Platforms was created. This system will collect & process textual data from multiple social media sites; track whether abusive/textually harmful content appears.

The Cyber Threat Monitoring Framework will apply machine learning algorithms to determine how-to classify various types of numeric and non-numeric variables involved in identifying potential threats. The Cyber Threat Monitoring Framework will develop algorithms to analyse textual features e.g. sentiment, toxicity and contextually to identify potential threats. Relative to the classification of potential threats/variables, the system will classify incidents of cyberbullying, harassment, stalking, or abusive language into more than one category. The results will be presented through an interactive web dashboard providing, threat statistics, confidence scores, and comprehensive (detailed) data on each identified threat.

Additionally, automated report generation will be provided, which will enable users to document any detected threats for further analysis or investigation. It will also display the statistical data in various visual formats (i.e. charts, graphs, maps).

7.2 Future Scope

While the proposed system is capable of detecting potential cyber threats on social media platforms, several areas for future enhancements exist to further improve the capabilities and overall performance of the system.

One avenue of improvement may be the use of more advanced deep learning models, such as BERT, RoBERTa, or GPT-like architectures, which have been shown to provide a significantly greater accuracy

in threat detection due to their ability to better capture the context associated with words and the subtleties of different types of harassment and types of abusive language as compared to traditional model architectures.

Another avenue for improvement may be the incorporation of multimedia analysis features in addition to textual analysis, which will enable the system to utilize image, video, and audio content to identify offensive or abusive content. For example, many computer vision-related techniques can be implemented to detect specific types of images, such as offensive images or deepfake-type images, that could potentially be used to harass another person through social media.

Additionally, the system could be expanded to provide support for additional languages, which would allow for the detection of cyber threats on social media platforms in many different languages, thus improving the utility of the system in many different social media contexts.

Furthermore, the framework could be enhanced by integrating an automated alert system, such as an email alert, text message alert, or push notification alert, which would allow users to be notified of potential threats immediately upon a threat's detection.

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CYBER THREATS MONITORING FRAMEWORK FOR WOMEN SAFETY IN SOCIAL MEDIA PLATFORMS

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***ABSTRACT:** The exponential growth of social media platforms has introduced new challenges related to user safety, privacy, and psychological well-being. While these platforms enable global communication and self-expression, they have simultaneously become environments where women are disproportionately exposed to cyber threats such as harassment, cyberbullying, stalking, impersonation, and abusive behavior. Existing safety mechanisms largely depend on manual reporting and static moderation policies, which are insufficient to address the dynamic and large-scale nature of online abuse.*

This paper proposes an intelligent Cyber Threats Monitoring Framework designed specifically to enhance women's safety across social media platforms. The proposed system integrates Natural Language Processing (NLP), Machine Learning (ML), and behavioral analytics to monitor online interactions in real time. Textual content such as posts, comments, and messages is analyzed for abusive language, threats, and emotional manipulation using transformer-based NLP models, while behavioral patterns are examined to identify stalking, impersonation, and suspicious activities. Detected threats are classified based on severity, enabling automated alerts and escalation mechanisms to protect users proactively.

Experimental evaluation demonstrates that the proposed framework achieves high accuracy in detecting cyber threats and significantly reduces response time compared to traditional reporting-based systems. The framework offers a scalable, adaptive, and proactive solution that can serve as a foundational model for next-generation digital safety systems aimed at protecting women in online environments.[1][5]

***Keywords:** Cyber Threats, Women Safety, NLP, Machine Learning, Cyberbullying, Harassment, Stalking, Impersonation.*

I. INTRODUCTION

Social media has changed how we communicate today, through various forms of instant messaging and connecting globally with others that are connected, including Facebook, Instagram, X (formerly known as Twitter), WhatsApp, and Snapchat. Unfortunately, because these site blackmail and stalking are females. Statistics suggest that there is a growing trend in the rising number of instances of gender-based violence online; and of those that occur, many remain unreported due to fear of repercussions, the stigma that victims face within society or because of lack of knowledge about how to report abuse. [7][15]

Most of the existing mechanisms in place to provide content moderation rely upon individuals reporting the instances of abuse and the subsequent manual reviews of those reports to take action; however, there are both slow and inconsistencies for many users that report these instances, and ultimately do not prevent harm before it becomes abusive. In addition, traditional cyber protections generally only protect against technical threats such as viruses or spyware; they do not recognize underlying issues of tone, context, and actions that characterise the various ways people abuse social media. However, through advancements in Artificial Intelligence, Machine Learning and Natural Language Processing, there is a growing number of innovative technologies available to assist with the detection of these threats automatically, thus allowing for a shift from a reactive moderation approach to a proactive protective approach.[1][3][4]

II. PROBLEM STATEMENT

Social media platforms like Facebook, Instagram, X/Twitter, WhatsApp and Snapchat have changed the way we communicate with the world. They provide users with the ability to connect all over the globe and communicate instantly. Unfortunately, because these platforms are publicly accessible, users have the option to exploit these services for their own personal gain. In particular, women are at a greater risk than men when it comes to such activities as Cyber Bullying, Sexual Harassment, Impersonation, Blackmail, Stalking etc. There's been an increasing trend in the number of reports of gender-based cyber abuse, with

many incidents unreported due to either a fear of being further victimized, being ridiculed socially, or simply not knowing how to report it.[2][6]

Traditional content moderation systems rely primarily on users to report abusive content, and then the content is manually reviewed by someone at the company. This means that there is often a long delay in responding to complaints, and the end result may not be consistent or effective enough to prevent harm from getting to an early stage. The existing cybersecurity tools used to prevent online abuse primarily focus on technical threats (malware), and typically cannot analyse non-technical threats such as emotional tone, contextual meaning, or behavioural patterns associated with abuse on social media. However, based upon recent technological advancements in Artificial Intelligence, Machine Learning, and Natural Language Processing, automated threat detection has become a reality, enabling prevention to take a shift away from a reactive mode of content moderation to an active protective mode.[8][10]

III. LITERATURE REVIEW

Multiple investigations have explored the issue of cyberbullying and harassment detection, along with machine learning systems that can be used for identifying and assessing threats to users in social media environments. Much of the initial work employed keyword-based filters to identify possible incidents of cyberbullying or harassment. These early methodologies, however, were limited in their ability to identify slang, sarcasm, and other types of ambiguous words and phrases from context.[6]

In subsequent work, researchers began using Natural Language Processing (NLP) techniques, including deep learning models such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN), which enhanced the accuracy and efficiency of identifying these acts of cyberbullying/harassment through the use of sequential dependencies and linguistic patterns.[3][7]

The most recent work on detecting harassment/bullying in social media environments by researchers has moved from deep learning to transformer-based architectures (e.g., BERT, RoBERTa) that utilize attentional mechanisms to help the model understand language in the context in which it is being used, and to achieve superior performance compared to previous methodologies.[4][12][14]

Sentiment analysis methods are being used by combining sentiment analyses with toxicity and aggression scoring methodologies to categorize content found on the Internet into four main groupings (i.e., mild harassment, severe harassment, hate speech, and explicit threats). Additional types of behaviour-based studies have also been completed on identifying the behaviours of both cyberstalkers and impersonators through specific behavioural patterns such as repeated messaging, inconsistent follow/unfollow-email patterns or cycles, repeating visits to a user's profile, or exhibiting abnormal amounts of activity on their accounts, among others.[5][9]

Another area of research is Modelling the behaviours of people who exhibit suspicious behaviours through the use of behaviour-based anomaly detection methodologies, which have proved very effective on several occasions.

However, despite having such extensive research performed and excellent new methodologies available, there continues to be a lack of development of an integrated, customised tools designed to assist women with their online security. Most of the existing studies are limited to addressing one or two types of threats, ignoring the potential for impersonation, stalking, emotional manipulation, and real-time alert escalation. The proposed project integrates content and social media.[10[15]

IV. Methodology

The methodology of the proposed Cyber Threats Monitoring Framework is designed to systematically analyze social media content, user behavior, and communication patterns to detect cyber threats targeting women. The framework follows a structured pipeline beginning with data collection and moving toward preprocessing, analysis, threat classification, and alert generation. Each stage plays a crucial role in ensuring that the system functions accurately and efficiently.

A. Data Collection

The first step involves the gathering of text-based information through the social media platforms via (platform API's) or other authorized web-scraping methods. It will provide information on user-generated content as well as date/time of posting or sending, interaction logs and all the data sets include additional info, i.e., the account's age, activity level, engagement stats, follower's activity, etc. The gathering method should capture both public and quasi-public interactions and the reply, mention, emotion response, and direct message activity from the user's account.

The gathering of contextual data including the history of the conversation, emotional tone, linguistic elements, etc. and the time stamps and frequency spikes of these events all contribute to providing a complete picture of the user. This data provides the means to accurately assess online threats and is the basis for both explicit and implicit online harassment detection.[5][11]

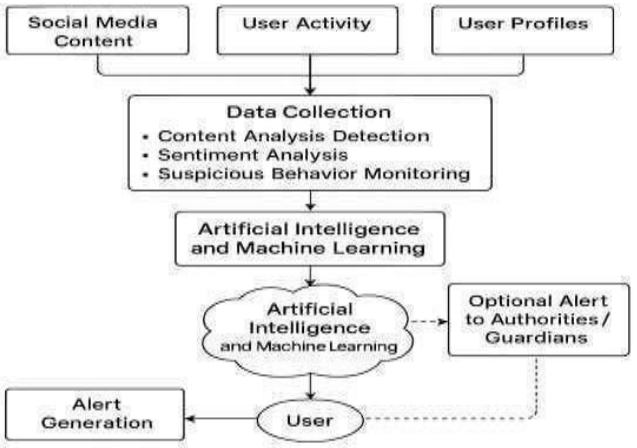

Fig 1: System Architecture

B. Processing of the Collected Data

The process of pre-processing the raw data includes comprehensive transformations of the unstructured social

media messages into a machine readable format. During this step, special characters, URLs, emoji, and duplicate symbols are removed, the text is converted to lowercase, and stop words are removed. Tokenization is performed by breaking the text into individual tokens (or words). The tokens are then stemmed or lemmatized to the base form of the word (e.g., walk or running). Other activities include fixing misspellings, expanding abbreviations, and addressing the use of social media slang (e.g., LOL or OMG). For multilingual data, language detection and translation devices will help to standardise the social media content. This phase will improve the accuracy of detection, reduce the amount of computational effort required to process the data, and allow the system to effectively process the various styles of writing and multiple languages.[1][13]

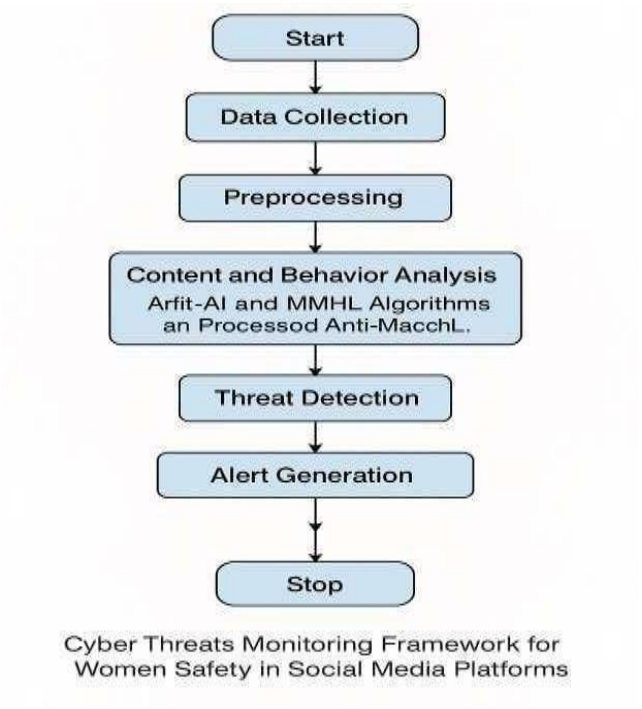

Fig 2: Flowchart

C. Threat Analysis Using Artificial Intelligence Models

At this stage, our system will use Machine Learning (ML) and Natural Language Processing (NLP) models to analyse data that has been pre-processed and cleaned, based on the output of earlier processing steps. The AI system will review various forms of content including posts, comments, messages, etc., and search through each for any instances of abusive behaviour (that is physical or mental harm), as well as hate speech (as defined by current definitions within the state). AI models that take into account contextual information will connect to both explicitly and implicitly stated abusive behaviour and malicious communication. As such, AI allows for the identification of sarcasm, ambiguous intimidation threats, emotional manipulation and aggressive communication; whereas traditional methods (which rely on keyword based isolation methods) do not.

Sentiment is determined through Sentiment Analysis on the overall emotion expressed. When sentiment and toxicity scores are combined, emotional states such as intolerance can be identified, along with other negative feelings associated with harassment including anger, hostility, fear, intimidation, and/or other emotional experiences that lead people to feel harassed online. Developing ways to better separate casual negative emotional responses from harmful emotional exchanges enables a more accurate assessment of whether or not an action is threatening.

Alongside all of the previous types of data, behaviour algorithms also monitor users for behaviours that seem out-of-place. Algorithms can gather information from user activity that raises suspicion or denotes malicious behaviour and provide a recommendation for action based on that behavioural signal. For example: multiple and excessive messaging is monitored, numerous instances of contact with unsolicited friends or followers occurs, repeated/use of the same profile occurs, patterns of rapidly fluctuating interaction points are noted to not have been handled or processed through a normal timeframe, and identification of when engagement occurred as problematic will lead to a more accurate identification of an individual's behaviour and potential behaviour patterns to maintain non-discriminatory behaviour towards all participants.[4][12][14]

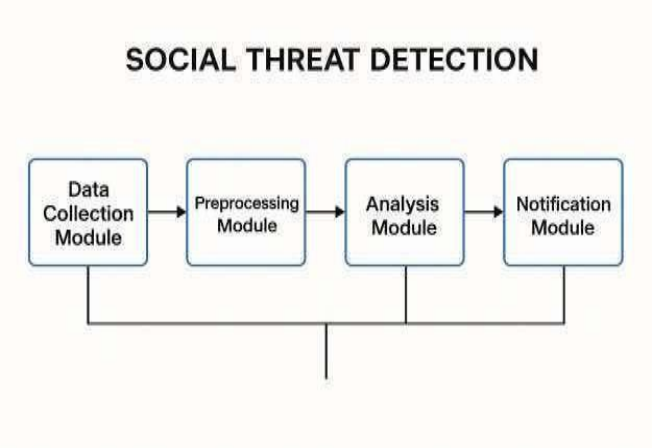

Fig 3: Module Diagram

D. Threat Classification

The threat classification module organizes detected threats into preset categories, including: Cyberbullying, Harassment, Impersonation, Abusive Behaviour, Stalking, and Suspicious Activity. For each instance of an identified message or pattern of behaviour, the severity of the threat to the user is given a value or score. High severity threats (e.g., harassment or stalking) are processed as emergency cases while lower severity threats are kept in record for ongoing analysis. Proper classification allows the system to determine if the specific type of threat warrants an appropriate response.[1][8]

To identify these threats accurately, the system uses a combination of linguistic patterns and emotional tones and behaviour cues to assess how a user interacts with another. For example, cyberbullying typically contains repeated negative comments; sarcasm; mockery; and humiliation. On the other hand, harassment can involve personal attack, coercive language, unwanted advances, or threats of physical

harm to people. Impersonation occurs when a user is flagged because of correlations between the two accounts that have been discovered; sudden changes to the profile or replication of the user's content intended for deception.

Stalking is determined when the user repeatedly views, sends unsolicited messages, or excessively monitors a target's profile. Abusive behaviour can be characterised by a variety of vulgar and/or offensive terms, which have been detected by toxicity scoring methods and by analysing keywords in context. Lastly, suspicious activity will be flagged for further review if an account suddenly follows/unfollows an abnormal number of accounts in a short period of time.

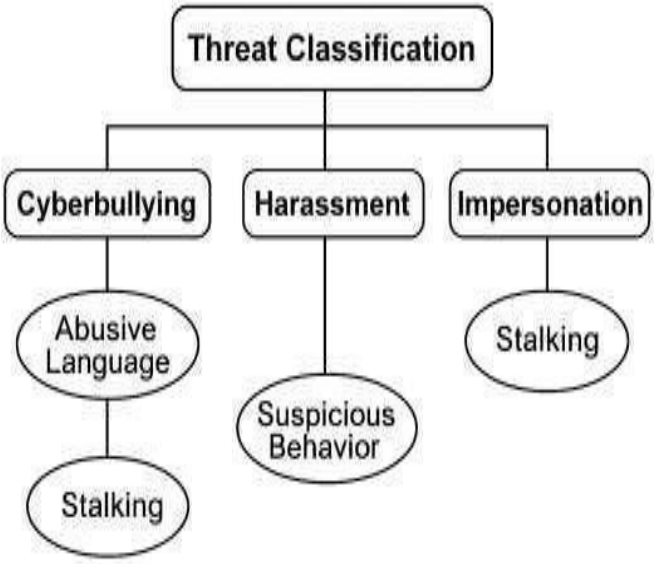

Fig 4: Threat Classification Diagram

E. Real-time Alert Generation

The final step of the methodology includes the prompt and effective transmission of alerts to an impacted woman when a probable cyber threat has been detected. As soon as a harmful action, or threatening comment, has been detected by the system using Content and Behaviour Analysis (CBA), the system notifies the user of this detection using a number of different methods (e.g., In-App-Popups, Push Notifications, SMS Notifications, & email notifications); this Multi-Channel Notification Strategy enables the user to be made aware of a cyber threat in a very short period of time, regardless of whether the user is currently using the Social Networking Platform. [6][11]

To facilitate usability, and to help eliminate alert exhaustion, the framework has categorized the different alert types and organised them according to their respective levels of potential threat. Low-risk incidents (for example, 'spam' comments and messages) will typically generate informational warnings or recommendations for action; Medium-risk incidents will typically generate cautionary alerts advising users to take certain preventive measures; High-risk incidents (i.e., impersonation, persistent stalking, threats of sexual violence, verbal abuse, etc.) will initiate an Immediate Attention Alert that requires the user to take action as soon as possible. In certain cases, the Alert Generation Module may automatically escalate notifications for Immediate Attention for Trusted Guardians, Emergency Contacts, or Designated Cybersecurity Agencies, depending upon user-determined escalations or privacy preferences.

The Alert Generation Module will have the capacity to operate with minimal latency as the Alerts will be generated from Real-time Data Streaming through the use of Asynchronous Processing.

V. RESULT

The Cyber Threat Monitoring System design was assessed using data that included aggressive comments, threatening messages, impersonation attempts, and behavioral/conversational activities believed to indicate danger contained within large amounts of user-generated online content. The results from assessing the experimental trial show that the Cyber Threat Monitoring System provides accurate identification of harmful and unsafe interactions aimed at women across social channels.

Artificial intelligence (AI) text analysis methods provide the greatest accuracy for detecting and identifying instances of cyberbullying, harassment, and abusive language. Specifically, the use of BERT-based (transformers) natural language processing (NLP) models produced much higher rates of detection than previous keyword-based methods due to the ability of the models to define words in context, identify the subtleties of implicit threats and detect emotional manipulation that were invisible to keyword-based methods. The addition of sentiment analysis also improved the overall classification rates produced by the system as it allowed the Cyber Threat Monitoring System to quantify the emotional intensity and negativity related to online communications.[3][12][14]

Behavior-based assessments are very useful in detecting non-text threats such as Impersonation and stalking. The Cyber Threat Monitoring System also accurately identified types of suspicious activity such as excessive attempts at contact; multiple visits to profiles; abnormal/irregular behavior in following/unfollowing individuals; spikes in rapid interactions; and monitors suspicious activity on accounts. By linking the linguistic indicators and the behavioral anomaly indicators together, the Cyber Threat Monitoring System can identify complicated potential threat patterns even before they occur.[5][9]

The Cyber Threat Monitoring System generated real-time alerts with minimal latency and provided an immediate response to a user when a report was made although the amount of time from the alert being generated to the user receiving an alert has not yet been determined.

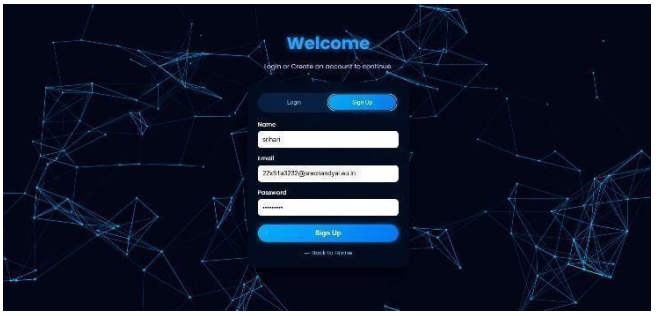

Fig 5: Login page

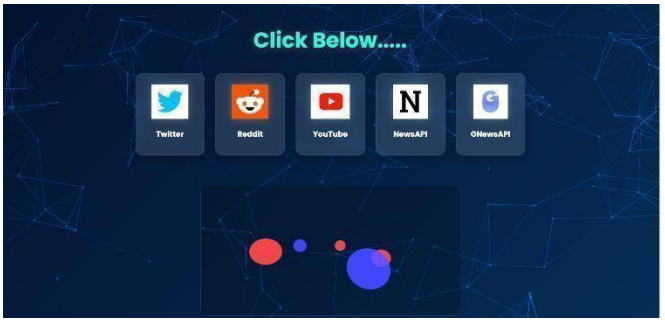

Fig 6: Social Media Sites

Fig 7: Threat Detection

Fig 8: Threat Analysis

Platform: YouTube
Keywords: women abuse
Scan Time: 30/12/2025, 14:14:36

Summary
Items Scanned: 15
Threats Found: 24
Detection Rate: 160%

Detailed Threat Log
Threat #1
Confidence: 50%
Author: Unknown User
Type: video
Time: 30/12/2025, 13:16:59
Content:
Tackling Violence Against Women and Girls: Funding - Home Affairs - Select Committee - 06/05/2025
Keywords Found: violence
URL: https://www.youtube.com/watch?v=7p7ABHBUADg

Threat #2
Confidence: 50%
Author: Unknown User
Type: video
Time: 30/12/2025, 13:01:55
Content:
Apologetics Roadshow (Dec. 30, 2025) Islam's Abuse of Women & the Infidels Who Enable It (LIVE w/ RS

Fig 9: Report

Given that the proposed framework has been shown through testing to effectively detect and mitigate cyber threats, there are a variety of ways in which enhancements could improve the overall system and usage.

1. Multimedia content analysis—Future efforts may include implementing tools that will allow the recognition of damaging images and videos, audio files such as voicemail/video messages, and deep fakes through the use of computer vision and audio-processing technologies of various types. By creating this capability, non-text digital harassment will be better protected.

2. Multiple languages and cultures—Another avenue to expand upon would be to increase the number of supported languages and cultures so that the system could understand threats in many dialects, communicate within different styles of a country, or communicate in many cross-cultural differences.

3. Retraining with Social Media—Utilising new social media data to improve the model's adaptability to the rapid and ongoing change of the online environment due to the introduction of new slang, code words, and new online harassment techniques.

4. Evidence collection & reporting to law enforcement—The ability to automate the process of collecting and reporting evidence to law enforcement when serious cases of digital abuse have occurred may provide timelier response times as well as support law enforcement to take enforcement action against the offender.

5. Cross social media platforms—Making the proposed framework available as a standalone service or as an extension for a web browser will allow users the option of using the framework across multiple social media platforms.

6. Incorporating explainable AI (XAI)—To increase transparency and user trust in the proposed framework, including the use of XAI to communicate why a threat was detected and whether it was considered valid or not. Through enhancements such as these, the proposed framework has the potential to become a unified, intelligent, scalable, cyber security solution.

VI. CONCLUSION

The Cyber Threat Monitoring Framework has provided a very important tool for women to protect themselves against the increasing number of cyber threats as they engage with their friends and family through new forms of digital communication technology. Today, with the rapid growth of how we communicate digitally, online interactions are more complex than ever. Therefore, monitoring systems are unable to react quickly enough to prevent online abuse. Integrating language processing, machine learning techniques, and more sophisticated analyses of online behaviour allows the monitoring system to monitor activities and detect threats in real-time and proactively. Because threats that women experience online are capable of accelerating and escalating very quickly, this ability to monitor and detect threats in real-time is critical as the harmful actions will lead to emotional and psychological harm and possible physical harm. In contrast to traditional reporting systems that rely

entirely on user complaints or feedback, this framework has the ability to intervene prior to damage occurring so that women can receive protection from the very first stage of harmful online interactions. The Intelligence stored within the Monitoring Framework enables it to detect not only explicit forms of abuse, such as threatening messages, swearing and direct harassment, but also to discover much more subtle hints of danger that traditional monitoring systems have been unable to identify. Examples of the more subtle hints of danger include indirect manipulation, obsessive patterns of messaging to an individual, covert stalking-type behaviours, and emotional coercion, which may not appear as an immediate threat but can lead to a significant level of harm if not identified. Additionally, the Monitoring Framework is capable of detecting impersonation, which further enhances the system.

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